Distribution, Technique, and the Transformation of Accumulation: Reconstructing Capacity Utilization in the Center and Periphery*

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August 6, 2026

Abstract

Capacity utilization has been treated either as a given macro ratio, an inherited benchmark, or an empirical residual. That leaves unresolved the problem of how accumulation is transformed into productive capacity under capitalist disequilibrium. This chapter reconstructs capacity utilization as a structurally mediated object, recovered through the identification of the transformation elasticity θ, where distribution mediates the conversion of accumulation into productive capacity. Applying this framework comparative-econometrically to the United States (1947–2024) and Chile (1920–2024), we demonstrate that the transformation elasticity is endogenously determined through the choice of technique and the balance of class forces. In the center, the shift from a wage-led Fordist nexus to a post-Fordist relative price channel led to institutional decoupling and overaccumulation. In the periphery, this transformation is structurally truncated by external balance constraints and balance of payments dynamics. Our findings challenge the view of utilization as a simple residual, showing it to be a key institutional marker of macro-political-economic reproduction.

Keywords: Capacity Utilization; Transformation Elasticity; Choice of Technique; Cointegration; Marx-Okishio; External Balance; Center-Periphery.
JEL Codes: B51; C32; E01; E22; E32; P16.

*This paper benefited from AI-assisted editing for syntax refinement, structural suggestions, and stylistic consistency checks. All analytical arguments, historical interpretations, and substantive content remain the author's sole responsibility.

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1 The Unresolved Problem of Capacity Utilization

A This paper examines how the rate of profit, capital accumulation, and productive capacity formation interact under conditions of distributional conflict between labor and capital.

B

[**PLACEHOLDER**] Section 1 is unwritten. What exists above is a single orienting sentence; the section that the locked outline requires does not yet exist. The remainder of this box states what it has to do, so the gap is legible rather than silent.

C

[**PLACEHOLDER**] **Burden of the section.** Establish that capacity utilization is unresolved in the treatments that have come down to us, in three distinct ways, and show that these are three faces of one problem rather than three separate difficulties. First, utilization treated as a *given macro ratio*: taken as observed when its denominator is not. Second, utilization treated as an *inherited benchmark*: anchored to an engineering or survey norm imported from outside the accounting framework that is supposed to give it meaning. Third, utilization treated as an *empirical residual*: whatever is left after output has been fitted to capital, which presupposes the very transformation relation at issue. The third case is where the filtering literature sits, and the objection to it here is structural rather than methodological — a filter that fits output to capital imposes a unit transformation elasticity by construction, and so conflates utilization with the object that would be needed to identify it. That argument is available from this chapter's own accounting framework and requires no external authority.

D

[**PLACEHOLDER**] **Bottom line up front.** Per the writing calibration in force, the contribution must be visible by the end of the second paragraph: what the paper examines, how it identifies the transformation elasticity, and what the comparison of the United States and Chile establishes. The reader should not have to reach Section 3 to learn what is being claimed.

E

[**PLACEHOLDER**] **What must not appear here.** Long literature catalogues. The full center–periphery discussion, which belongs in Section 5. Estimator debates, which belong in Section 4 and Appendix C. Detailed methodological caveats. The section motivates a problem; it does not litigate the solution.

F

[**PLACEHOLDER**] **Closing-line burden (locked).** The section must arrive at: *the utilization problem cannot be resolved at the level of the ratio alone; it must be reconstructed through the relation that governs productive-capacity formation.* This hands off directly to Section 2.

G

[**PLACEHOLDER**] **Also unresolved: the abstract's sample window.** The abstract states a United States window of 1947–2024, while Sections 4 through 6 estimate over 1929–2024 with MATHPLACEHOLDER0XYZ. The two cannot both be right. Note further that 1929–2024 spans 96 years against a reported MATHPLACEHOLDER1XYZ of 94, and that the estimation prose elsewhere gives the effective span as 1931–2024 after lag-1 conditioning and listwise deletion — which is consistent with MATHPLACEHOLDER2XYZ and suggests the table titles, not the sample, are what needs correcting.

2 The Micro-Meso Hinge: Okishio, Vidal, and Capacity Formation

A Chapter 1 established that the macroeconomic capacity ceiling is not a neutral engineering constraint, but a historically contingent outcome conditioned by the balance of class power. Yet, identifying this distributional conditioning at the macro level leaves the underlying transmission mechanism black-boxed. How, precisely, does distributive conflict physically alter the conversion of accumulated capital into productive capacity?

B Answering this requires dismantling a hidden ontological premise of heterodox growth models: the unified firm. Whether treating the normal rate of capacity utilization ($\mu_t \equiv Y_t/Y_t^p$, where Y_t is realized output and Y_t^p is potential capacity) as an endogenous behavioral target or an exogenous gravitational anchor, standard frameworks presume a coherent organization capable of selecting a benchmark, enforcing it across the labor process, and sustaining it over time.

C This premise collapses when confronted with the social organization of capitalist production. In a decentralized capitalist economy, individual capitals commit fixed investments without *ex-ante* coordination, confronting one another in competitive markets. Within the enterprise, production decisions encounter contradictory managerial and worker imperatives. Capacity utilization cannot be reduced to a target selected by a single optimizing entity. To theorize the political economy of productive capacity, we must replace the unified firm with two interconnected frameworks: Okishio's theory of decentralized market anarchy, and Vidal's theory of the internally divided firm.

2.1 Smooth Reproduction and the Unified-Firm Premise

A Smooth capitalist reproduction requires strict material proportionality between departments of production, a condition formally proven by Basu (2022) and defined by Okishio (2022) as the equilibrium accumulation trajectory. Bergmann et al. (2009) refer to this balanced trajectory—where realized output meets productive capacity ($\mu_t = 1$)—as the "golden path." However, this proportional benchmark serves strictly as an analytical condition of feasibility, not a historical tendency or a gravitational attractor. While a utilization rate of unity ($\mu_t = 1$) is a necessary condition for balanced reproduction, it is not sufficient; full macroeconomic reproduction also requires that the rate of accumulation matches the growth rate of the labor force and technical change. Nothing in the logic of decentralized accumulation establishes that competition will reproduce these proportions or pull the macroeconomy back toward unity.

B Because private investment decisions are made without *ex-ante* coordination, real capitalist economies routinely violate the strict conditions of the golden path. Okishio (2022) demonstrates that this chronic disproportionality is not a temporary friction but the structural norm of decentralized accumulation. Treating this transient benchmark as a behavioral target relies on the flawed premise of the unified firm.

C In reality, the denominator of the utilization ratio (productive capacity Y_t^p) is not an engineering constant, but a contested settlement of divided management and shop-floor struggle; the numerator (realized output Y_t) is similarly hostage to the anarchy of demand. Smooth reproduction supplies a reference case against which this historical non-coincidence can be studied; it does not describe the path capitalist accumulation actually follows.

2.2 Decentralized Accumulation and Market Anarchy

A If the "golden path" of smooth reproduction is merely a benchmark of feasibility, the actual trajectory of capitalist accumulation is governed by the decentralized organization of investment. Capitalist production is coordinated through exchange, meaning that the social validation of private decisions occurs only *ex-post*. Because money mediates exchange, sale and purchase are separated moments of the capital circuit (Basu, 2022). This constitutes the logic of capital as impersonal domination: capitalists make their production and investment decisions with the law of value operating *"behind their backs"* [1]. They are subject to market forces and value relations they neither control nor consciously direct. Okishio (2022) locates the source of macroeconomic instability precisely in this structural anarchy. Individual capitals commit to fixed investment on the basis of private profit expectations, yet the market validates those commitments only after production has begun. No firm can know in advance whether its expansion fits the investment and consumption decisions taken elsewhere. The coordination problem is not one of imperfect information surrounding an otherwise coherent plan; no economy-wide plan exists.

B Smooth reproduction requires proportional expansion between Department I, which produces means of production, and Department II, which produces consumption goods. Because private investment does not enforce this proportionality, firms extrapolate past prices, profit rates, and sales. A signal that induces expansion in one branch routinely produces excess capacity before complementary demand materializes elsewhere. Contraction works through the same cumulative mechanism: falling sales cut investment, reducing aggregate demand and validating further cuts. Prices register the resulting imbalance, but they do not prevent the sunk commitments that produced it.

C Fixed capital gives this temporal lag its structural force. Structures and machinery embody long-horizon commitments that cannot be reversed when relative prices or demand conditions shift. Once installed, an unvalidated project leaves the firm to operate below its anticipated scale, defer maintenance, hold equipment idle, or devalue the asset. These responses distribute the costs of market anarchy among firms, workers, and creditors, but they do not restore the missing *ex-ante* coordination.

D Idle capacity is therefore a recurrent, structural form of decentralized accumulation rather than a temporary friction. Private investment physically builds capacity before the market can validate its social use, repeatedly generating a gap between installed productive potential (Y_t^p) and realized output (Y_t). In this framework, the capacity utilization rate (μ_t) operates as the macroeconomic "exhaust valve" for structural disproportionality. Rather than adjusting investment to smoothly restore the golden path of balanced growth, the system absorbs the shock of uncoordinated accumulation through fluctuating utilization rates. The condition $\mu_t = 1$ is therefore at most a transient point of accidental proportionality, while persistent divergence records the ongoing reality of market anarchy.

E Yet Okishio's macro framework leaves a crucial gap: it does not explain how firms translate this external anarchy into internal operating decisions. A firm does not receive a price signal and

[1] Karl Marx, *Capital*, Volume I, Chapter 1, Section 4 ("The Fetishism of Commodities and the Secret thereof"), where he describes the law of value operating as an overriding law of nature when private producers exchange products.

respond as a frictionless, unified unit. It processes the signal through an organization divided over control, cooperation, effort, and the allocation of resources. Vidal's internally divided firm provides that missing mediation.

2.3 The Internally Divided Firm

A Vidal reconstructs Marx's Volume I theory of the labor process for the contemporary organizational era. Marx established that capitalist management faces a constitutive contradiction: it must simultaneously organize the cooperative labor required for production and enforce the coercive control required for valorization (surplus extraction). Vidal demonstrates how this contradiction fractures the modern capitalist firm, treating it as a divided social organization rather than a single-minded production function (Vidal, 2013, 2022). Management must extract surplus value through control over labor, but production also requires workers' cooperation, tacit knowledge, and problem-solving capacity. Standardization and surveillance secure a measurable pace of work while weakening the discretion needed to sustain complex production. The firm cannot maximize both imperatives at once because the labor process that produces value is also the site where workers contest its extraction.

B Workers respond to this contradiction by actively calibrating effort, cooperation, machine pacing, and the deployment of tacit knowledge. Capitalist management extracts potential capacity through concrete shop-floor mechanisms: increasing machine speed (speedups), deferring maintenance to hit short-term throughput targets, and enforcing rigid surveillance. Workers contest this extraction through labor turnover, collective exit, or the deliberate withdrawal of problem-solving cooperation and effort. Throughput, quality, and maintenance therefore depend on social relations that no engineering specification fixes in advance. A production target that raises machine speed by deferring maintenance may increase current output while degrading sustainable capacity. A directive that suppresses worker discretion may meet a short-term labor-control objective while worsening workflow bottlenecks. Labor effort is consequently not a fixed technical parameter applied to a pre-existing capital stock; it is an active, contested determinant of the capacity denominator itself.

C This dynamic enforces a strict wedge between *engineering capacity* and *practical capacity*. Engineering capacity records the theoretical maximum output of equipment under ideal technical conditions. Practical capacity records the sustainable output rate dictated by actual maintenance schedules, product-mix complexity, and the negotiated intensity of labor. When managers report capacity to national statistical agencies, they rely on conventions shaped by their organizational position and the immediate pressures of the shop floor. Aggregate capacity statistics therefore do not reveal a neutral physical ceiling; they aggregate situated, institutional judgments. For the purposes of macroeconomic modeling, the productive capacity ceiling (Y_t^p) is not an exogenous engineering constraint, but an endogenous settlement of workplace conflict.

D Because the valorization-labor process contradiction cannot be solved, firms settle into organizational configurations of "mediocre sufficiency" (Vidal, 2022). These are hybrid arrangements that reproduce production without resolving the underlying conflict between control and cooperation. Partial worker autonomy coexists with rigid discipline; continuous improvement programs coexist with brutal throughput pressure; idle machinery coexists with binding bottlenecks. These arrangements are not temporary frictions or departures from an otherwise unified, optimizing firm.

They are the permanent, structural forms through which capitalist management physically converts accumulated capital into usable capacity.

2.4 Distribution, Technique, and the Transformation of Accumulation

A The choice of technique connects the divided firm to distribution, governing how accumulated capital is physically transformed into productive capacity. Capitalists adopt a new technique when it reduces costs at prevailing prices and real wages. Okishio's theorem shows that a viable capital-using, labor-saving (CU-LS) technique raises the profit rate when the real wage remains fixed (Okishio, 1961). Marx's account allows the same form of technical change to depress profitability when rising capital intensity and the distribution of productivity gains increase the capital advanced relative to surplus value. The Marx–Okishio threshold identifies the rate of real-wage growth that separates these outcomes (Basu, 2022). It is a distributive boundary set by the balance of class forces, not a technical constant.

B This boundary gives wage pressure a precise role in mechanization. When organized labor compels real wages to follow productivity, labor-saving techniques become more attractive while workers retain part of the resulting productivity gain. Under the institutional settlement of Atlantic Fordism—what Basu (2026) formalizes as the "strong labor" closure—this mechanism joined rapid mechanization to continuing conflict over profitability (Vidal, 2015, 2019). Wage-led mechanization was not a harmonious productivity bargain; the same distributive pressure that induced technical change contested capital's claim on its proceeds.

C The weakening of organized labor after the Fordist crisis altered that mechanism. Under the "weak labor" Post-Fordism, wage suppression reduced the distributive pressure to replace labor, even as lower relative machinery prices and technical change in capital-goods production continued to favor mechanization. Distribution and machinery prices therefore constitute a **dual-channel mechanism** in the choice of technique. Their relative force changes across accumulation regimes, which prevents the wage share from being treated as an external control added to an otherwise autonomous technical relation.

D These channels still operate through the contradictions identified above. Firms choose techniques before demand validates the associated capacity, and they operate those techniques through labor processes that divide control from productive cooperation. Investment consequently does not translate one-for-one into a stable productive ceiling. The transformation depends on the composition of capital, the distributive conditions governing mechanization, and the organization through which installed equipment becomes usable capacity.

E Okishio and Vidal thus rule out a direct passage from accumulation to productive capacity. Decentralized investment generates disproportionality across firms and sectors; the internally divided firm transmits that disproportionality into contested operating conditions. The relevant object is therefore the aggregate **transformation elasticity ($\theta \equiv dy^p/dk$)**, defined in log-levels to ensure a dimensionless parameter, which governs how the accumulation of capital translates into potential capacity expansion under distribution and market anarchy.

2.5 Historical Regimes of Capacity Formation in the Center and the Periphery

A The theoretical framework developed above—combining Okishio's market anarchy, Vidal's divided firm, and the Marx-Okishio threshold—provides the political economy foundation for understanding how the transformation elasticity (θ) shifts across distinct historical eras. Mapping the dual-channel mechanism of technical change onto the institutional history of center and peripheral economies yields three core hypotheses regarding the structural evolution of capacity formation:

B Hypothesis 1: The Fordist Distributive-Mechanization Nexus in the Center (1947–1973) Under the institutional settlement of Atlantic Fordism, rising wage shares acted as the primary catalyst for intensive mechanization. Capitalists responded to sustained distributive pressure by aggressively substituting machinery for labor. This wage-induced choice of technique was structurally accommodated by institutionally coupled capacity planning: extensive plant structures and intensive machinery expanded in tandem with the new technical composition of capital. This regime was structurally stabilized by the mid-1950s—leading to a key econometric break in **1956** characterized by post-Korean War defense budget stabilizers, long-horizon corporate planning, and labor union bureaucratization. These institutions locked in capacity designs but began to restrict capacity utilization flexibility, sustaining productive capacity formation until the late-Fordist profit squeeze (circa 1968–1973) exhausted the regime's mediating capacity, triggering a systemic crisis of accumulation.

C Hypothesis 2: Post-Fordist Institutional Decoupling and Unproductive Overaccumulation in the Center (1974–Present) Under Post-Fordism, the suppression of labor's bargaining power neutralized the distributive driver of mechanization. This transition was triggered by the structural breaks of **1974** (the Bretton Woods collapse, OPEC oil shock, and severe recession) and **1980** (the Volcker interest rate shock and Reagan's anti-labor counter-offensive). These shocks permanently depressed the baseline level of capacity validation by establishing a neoliberal regime of wage repression and prioritizing short-horizon, financialized asset management over physical capacity utilization. While aggregate capital accumulation continued, it fundamentally decoupled from structural plant expansion. In the absence of wage-induced productive innovation, the *relative price channel* (the cheapening of short-horizon equipment and ICT) dominated the choice of technique. Capital was increasingly allocated to non-productive, financially reversible processes rather than long-lived productive capacities. This institutional decoupling generated a chronic capacity realization failure, resulting in a severe structural overaccumulation regime where marginal capital growth yields diminishing or negative potential capacity expansion.

D Hypothesis 3: Peripheral Subordination and the Balance of Payments Truncation (1920–Present) In the capitalist periphery, the transformation of accumulation into productive capacity is structurally truncated by external financial subordination and the Balance of Payments (BoP) constraint. Unlike the center, where domestic class compromise or financialization dictates the choice of technique, peripheral mechanization relies heavily on imported capital goods and technologically non-substitutable inputs. When external financing compressions bind—whether under the inward-looking state-led regimes of the mid-century or the financialized neoliberal export models of the post-Fordist era—foreign exchange scarcity generates a hard realization bottleneck. The effective profit share available for reinvestment is squeezed, blocking the materialization of accounting surplus into productive capacity. This external subordination forces a chronic

peripheral accumulation wedge: nominal profitability coexists with truncated mechanization and stalled structural envelopes. Consequently, the peripheral productive ceiling is locked between domestic distributive struggles and the disciplinary regime of global financial markets, permanently destabilizing the long-run coupling between plant and machinery.

E These hypotheses establish that the productive ceiling is not a neutral engineering constraint, but a contingent outcome of decentralized accumulation, the balance of class power, and a country's geopolitical position within the world system. The immediate task is to translate these historical-institutional claims into a formal mathematical architecture. Section 3 will therefore formalize the dual-channel choice of technique, the heterogeneous composition of capital, and the peripheral foreign-exchange bottleneck to derive the regime-specific transformation elasticity (θ) that governs the passage from accumulation to productive capacity.

3 The Hinge Object: θ and the Conceptual Framework

3.1 The Baseline Model of Cost-Minimizing Mechanization (Homogeneous Capital Stock)

A Before formalizing the cost-minimizing choice of technique, we establish the descriptive macro context for the United States across five distinct historical windows: the pre-war Fordist recovery (1931–1941), the World War II mobilization block (1942–1945), the postwar Fordist Golden Age (1947–1973), the post-Fordist restructuring era prior to the Global Financial Crisis (1974–2007), and the post-GFC stagnation period (2008–2024). We track the average annual growth rates (mean annual log differences) of real corporate output alongside gross capital stocks built using the Gross Perpetual Inventory Method (GPIM), distinguishing private machinery and equipment (K^{ME}), private nonresidential structures (K^{NRC}), and public infrastructure in the form of government transportation, highways, and streets (K^{GT}). We report both Non-Financial Corporate sector Gross Value Added (y_t) and total economy Real GDP (y_{total}) to capture corporate-level dynamics relative to aggregate activity.

Table 1: Descriptive Macro Indicators for the United States (Growth Rates, % per year)

Variable	Pre-war Fordism (1931–1941)	WWII Block (1942–1945)	Postwar Fordism (1947–1973)	Post-Fordism Pre-GFC (1974–2007)	Post-GFC (2008–2024)
Real Corporate GVA (y_t)	5.80%	4.36%	4.98%	3.55%	1.99%
Real GDP (Total Economy, y_{total})	3.94%	9.92%	3.76%	2.97%	1.95%
Gross Machinery & Equipment (K^{ME})	1.37%	4.73%	8.49%	6.58%	3.66%
Gross Nonresidential Structures (K^{NRC})	1.08%	3.79%	6.38%	7.02%	3.99%
Gross Government Transportation (K^{GT})	4.07%	-0.14%	4.44%	2.05%	1.06%

B These indicators show the structural breaks defining each regime. During the pre-war recovery (1931–1941), corporate GVA grew at a rapid 5.80% per year. Public investment drove this, with infrastructure expanding at 4.07% while private accumulation lagged. World War II changed everything. The mobilization block (1942–1945) redirected capital toward war production. Civilian infrastructure contracted at -0.14% per year, while military spending pushed total GDP growth to a massive 9.92%. The postwar Golden Age (1947–1973) represents the historical peak of coupled accumulation. Machinery grew at 8.49% annually, structures at 6.38%, and public infrastructure at 4.44%. This balanced growth sustained a 4.98% average expansion of real corporate GVA. In the post-Fordist era, this coordination broke down. Public infrastructure growth fell to 2.05% before the GFC, and sank to a mere 1.06% after 2008. Meanwhile, private structures decoupled, growing at 7.02% pre-GFC before dropping to 3.99% post-GFC. This combination of private overaccumulation and public decay explains the long-run slowdown of corporate capacity formation.

C Following heterodox political economy, we represent actual aggregate output using a Leontief production function with fixed technical coefficients:

$$Y = \min\left(AL,\ \mu BK\right)$$

D where $A = Y/L$ is actual labor productivity, $\mu = Y/Y^p$ is capacity utilization, and $B = Y^p/K$ is maximum capital productivity. This production structure directly governs the rate of profit (r) through the fundamental heterodox accounting identity:

$$r = \mu B(1 - \omega)$$

E where ω is the wage share. Standard macroeconomic models frequently treat the structural coefficient B as an exogenous technological given, while treating the utilization rate μ as a behavioral constant smoothly targeted by a unified firm. We explicitly reject these assumptions. Instead, we treat the components of the profit rate identity as endogenous outcomes of class struggle and decentralized market anarchy.

F This section formalizes how distributive pressure forces cost-minimizing firms to alter their tactical choice of technique, endogenously determining the structural capacity coefficient B.

3.1.1 Tactical Choice of Technique and Induced Innovation

A Firms select a mechanization growth rate q to maximize the net productivity surplus. We define a as the growth rate of labor productivity and ω as the wage share. The effective distributive cost of deploying mechanization growth is therefore $q(1 - \omega)$. The firm solves the following tactical optimization problem:

$$\max_{q}\ c(q) = a - q(1 - \omega)$$

B This objective captures the induced-innovation logic: firms accelerate mechanization to expand labor productivity growth (a), but the distributive cost of capital intensity $(1 - \omega)$ holds them back. A technological frontier bounds this choice:

$$a = \Phi(q) = \lambda_0 + \lambda_1 q + \lambda_2 q^2$$

C The frontier parameters govern the structural limits of technical change. The base productivity growth $\lambda_0 > 0$ sets the floor without advanced machinery. The linear coefficient $\lambda_1 > 0$ captures the direct productivity payoff of mechanization. We impose the concavity condition $\lambda_2 < 0$ to ensure diminishing marginal returns to mechanization.

D Substituting the technological frontier into the objective function yields the unconstrained tactical problem:

$$c(q) = (\lambda_0 + \lambda_1 q + \lambda_2 q^2) - q(1 - \omega)$$

E Factoring out q gives the final objective function in terms of mechanization growth:

$$c(q) = \lambda_0 + (\lambda_1 - (1 - \omega))q + \lambda_2 q^2$$

F Differentiating with respect to q and setting the result to zero yields the first-order condition for optimal mechanization growth:

$$\frac{dc}{dq} = \lambda_1 - (1 - \omega) + 2\lambda_2 q^* = 0$$

G Solving for the optimal mechanization growth q^* gives:

$$q^*(\omega) = \frac{(1 - \omega) - \lambda_1}{2\lambda_2}$$

H The strict concavity of the frontier ($c''(q) = 2\lambda_2 < 0$) guarantees a unique and stable maximum. Differentiating q^* with respect to the wage share ω formally establishes the induced innovation mechanism:

$$\frac{\partial q^*}{\partial \omega} = -\frac{1}{2\lambda_2} > 0$$

I A rise in the wage share ($\omega \uparrow$) forces cost-minimizing firms to substitute machinery for labor. Rising wages drive mechanization directly through the first-order condition, pushing optimal capital intensity growth q^* upward.

3.1.2 Endogenous Transformation Elasticity

A Substituting the optimal technique back into the technological frontier recovers the normal growth rate of labor productivity under smooth reproduction:

$$a^N(\omega) = \lambda_0 + \lambda_1 q^*(\omega) + \lambda_2 [q^*(\omega)]^2$$

B This optimal technique dictates the structural trajectory of the Leontief parameter B. By definition, maximum capital productivity is the ratio of potential output to capital ($B = Y^p/K$). Expressed in dynamic growth rates, the growth of capital productivity (b) is the difference between capacity growth and capital accumulation ($b = g_{Y^p} - g_K$). Equivalently, it is the gap between labor productivity growth and mechanization growth ($b = a - q$).

C Evaluated at the smooth reproduction benchmark, this optimal gap is $b^N(\omega) = a^N(\omega) - q^*(\omega)$.

D We define the transformation elasticity (θ) as the structural parameter that translates capital accumulation into productive capacity growth ($g_{Y^p} = \theta g_K$). Substituting this definition into the growth rate of capital productivity ($b = \theta g_K - g_K$) reveals the fundamental mathematical link between the level coefficient B and the elasticity θ:

$$\theta = 1 + \frac{b}{g_K}$$

E To evaluate this elasticity, we must anchor the optimal technique to a baseline rate of capital accumulation. Following the normal reproduction path, the normal rate of capital accumulation (g_K^N) is driven by the normal profit rate (r^N) and the normal recapitalization share (χ^N), net of depreciation (δ):

$$g_K^N = \chi^N r^N - \delta$$

F Expressed dynamically relative to this normal accumulation path, the aggregate transformation elasticity becomes:

$$\theta(\omega) = 1 + \frac{b^N(\omega)}{g_K^N} = 1 + \frac{a^N(\omega) - q^*(\omega)}{g_K^N}$$

G This proves a crucial point: the transformation elasticity is not an exogenous technical constant. It emerges endogenously from the cost-minimizing choice of technique. When distributive pressure forces firms to alter mechanization, it simultaneously shifts the structural efficiency of accumulation.

3.1.3 Empirical Identification Strategy of the Baseline State-Dependent Elasticity

A The theoretical model derived in Sections 3.1.1 and 3.1.2 establishes that the transformation elasticity θ is an endogenous function of distributive pressure (ω). To bridge this formal analytical model with empirical data, we translate the continuous growth relation into an estimable long-run cointegrating specification.

B By definition, the aggregate transformation elasticity governs the marginal response of potential log output ($y_t^p = \ln Y_t^p$) to log real capital stock ($k_t = \ln K_t$):

$$g_{Y^p} = \theta_t g_K \quad \Longrightarrow \quad \frac{dy_t^p}{dk_t} = \theta_t$$

C Linearizing the theoretical function $\theta(\omega)$ around the initial state yields the baseline state-dependent elasticity specification:

$$\theta_t = \theta_0 + \theta_1 \omega_{t-1}$$

D where θ_0 is the baseline transformation elasticity of capital at the initial state, and $\theta_1 = \frac{\partial \theta_t}{\partial \omega_{t-1}}$ measures the marginal shift in capital efficiency induced by distributive class pressure. We condition the wage share at lag 1 (ω_{t-1}) to reflect the institutional lag between distributive bargaining and corporate capital expenditure execution.

E Integrating the differential relation $dy_t^p = (\theta_0 + \theta_1 \omega_{t-1}) dk_t$ over time yields the log-level potential capacity equation:

$$y_t^p = \alpha + \theta_0 k_t + \theta_1 (k_t \cdot \omega_{t-1})$$

F where α is a constant of integration. Adding a stationary stochastic error term u_t under the null hypothesis of cointegration yields the estimating equation:

$$y_t = \alpha + \theta_0 k_t + \theta_1 (k_t \cdot \omega_{t-1}) + u_t$$

G Capacity Recovery and the Residual Critique. Capacity utilization (μ_t) is not the regression residual, nor a detrended output gap. Following the locked capacity level anchor pinch year protocol, μ_t is recovered by applying the estimated baseline state-dependent elasticity $\hat{\theta}_t$ to capital growth ($g_{K,t}$), anchoring the level at the pinch year ($t^*_{US} = 1973$), and extrapolating against aggregate real output ($Y_t = \text{GDP}_t$). The full four-step sequence is executed in Section 4.2.

H Testable Hypothesis. This baseline state-dependent specification yields a testable empirical proposition:

- **Hypothesis 1 (The Okishio Distributive Dividend):** Domestic wage pressure compels cost-minimizing firms to substitute machinery for labor along the Mechanization Productive Frontier. The interaction coefficient between log capital and the lagged wage share is strictly positive:

  $$\hat{\theta}_1 > 0$$

  If $\hat{\theta}_1 > 0$, distributive pressure enhances the capacity-generating efficiency of capital, confirming that mechanization is endogenously induced by class struggle rather than driven by exogenous technical shocks.

I Section 3.1 closes the analytical identification of the homogeneous capital model. Translating the capacity equation into an estimable cointegrating specification — regressor orthogonalization, integration-order screening, and non-standard fixed-b inference — is executed in Section 4.

3.2 Heterogeneous Capital, the Mechanization Gap, and State-Dependent Viability

A Mainstream macroeconomics treats the capital stock as a homogeneous aggregate, erasing the internal architecture of the firm. Following Vidal's (2015, 2019) theory of the divided firm, we unbundle aggregate productive capital into extensive Nonresidential Structures (K^{NRC}) and intensive Machinery (K^{ME}). These components do not enter the labor process symmetrically. Structures form the passive infrastructure envelope—the factory floor and logistics shell. Machinery is the active, revolutionary force that restructures the production of relative surplus value (Marx, Vol. I, Ch. 15).

3.2.1 The Extensive Envelope and the Intensive Margin

A Nonresidential structures (K^{NRC}) define the **extensive margin** of accumulation. By construction, K^{NRC} evolves independently of short-run distributive conflict (ω) and anchors the mechanization productive frontier. It follows a normal accumulation corridor governed by long-run reproduction conditions and institutional closure:

$$g^N_{K^{NRC},t} = \chi^N r^N_t - \delta$$

B This extensive envelope $\Gamma(K^{NRC})$ bounds the space of tactical technique choice.

C Given this fixed infrastructural envelope, the **intensive margin** of accumulation is the tactical choice of the mechanization gap, defined in natural logs as:

$$\kappa_t = k^{ME}_t - k^{NRC}_t$$

D This gap measures mechanization intensity relative to the extensive plant scale. The capacity identity decomposes potential output into the extensive envelope and the intensive capacity yield Ψ_t:

$$y^p_t = \theta_{NRC} k^{NRC}_t + \Psi_t(\kappa_t, \omega_{t-1}, p^\kappa_t)$$

E Ψ_t is not a fixed engineering constant. It is the outcome of a decentralized cost-minimization problem governed by the regime of accumulation.

3.2.2 Cost-Surplus Maximization and the Okishio Viability Yield

A The firm chooses the mechanization gap κ_t to maximize the net viability surplus, subject to the regime-conditioned mechanization frontier $\Phi(\kappa_t) = \lambda_0 + \lambda_1\kappa_t + \lambda_2\kappa_t^2$.

B However, the *realized* capacity yield is governed by the **Marx-Okishio Viability Condition**: a cost-minimizing firm deploys a new technique only if it reduces unit costs at current distributive and price parameters. We formalize this as a cost-surplus maximization problem. The firm maximizes the gross engineering yield of the gap minus the effective distributive and price costs:

$$\max_{\kappa_t} \; c(\kappa_t) = \Phi(\kappa_t) - \kappa_t \cdot MC(\omega_{t-1}, p_t^\kappa)$$

C where $MC(\omega_{t-1}, p_t^\kappa)$ is the marginal viability cost of the gap. This cost is determined by two channels:

1. **The Distributive Dividend (Labor Saving):** Increasing κ_t substitutes machinery for labor. The value of this displacement is strictly proportional to the lagged wage share (ω_{t-1}), reflecting the wage settlement period prior to current capacity realization. A higher wage share *reduces* the net cost of mechanization, as the firm saves more on wages.

2. **The Relative Price Penalty (Capital Cost):** Increasing κ_t requires accumulating machinery faster than structures. The cost of this composition shift is proportional to the log relative price of machinery ($p_t^\kappa = \ln P_t^{ME} - \ln P_t^{NRC}$). A higher relative price *increases* the marginal cost of the gap.

D We specify the marginal viability cost as:

$$MC(\omega_{t-1}, p_t^\kappa) = \gamma_0 - \gamma_1 \omega_{t-1} + \gamma_2 p_t^\kappa$$

E where $\gamma_1 > 0$ captures the wage-savings dividend and $\gamma_2 > 0$ captures the relative price penalty. Substituting the frontier and the cost function into the objective yields the unconstrained tactical problem:

$$c(\kappa_t) = \lambda_0 + \left[\lambda_1 - (\gamma_0 - \gamma_1 \omega_{t-1} + \gamma_2 p_t^\kappa)\right] \kappa_t + \lambda_2 \kappa_t^2$$

F **First-Order Condition:** Differentiating with respect to κ_t and setting to zero:

$$\frac{dc}{d\kappa_t} = \lambda_1 - \gamma_0 + \gamma_1 \omega_{t-1} - \gamma_2 p_t^\kappa + 2\lambda_2 \kappa_t^* = 0$$

G **Optimal Mechanization Gap:**

$$\kappa_t^*(\omega_{t-1}, p_t^\kappa) = \frac{(\gamma_0 - \lambda_1) - \gamma_1 \omega_{t-1} + \gamma_2 p_t^\kappa}{2\lambda_2}$$

H Strict concavity ($2\lambda_2 < 0$) guarantees a unique maximum. The comparative statics establish the induced innovation mechanism:

$$\frac{\partial \kappa_t^*}{\partial \omega_{t-1}} = -\frac{\gamma_1}{2\lambda_2} > 0$$

$$\frac{\partial \kappa_t^*}{\partial p_t^\kappa} = \frac{\gamma_2}{2\lambda_2} < 0$$

I A rising wage share forces cost-minimizing firms to widen the mechanization gap, substituting machinery for labor. Conversely, a rising relative price of machinery suppresses the optimal gap, as the structural cost of the composition shift penalizes the mechanization program.

3.2.3 The Marginal Capacity Payoff and the Econometric Bridge

A The macroeconomic mapping from the mechanization gap to potential capacity is given by the **Marginal Capacity Payoff** (MP_κ), which is the derivative of the intensive capacity yield Ψ_t with respect to the gap κ_t.

B By the envelope theorem, the marginal capacity payoff is exactly the net marginal viability yield evaluated at the optimum:

$$MP_\kappa = \frac{\partial \Psi_t}{\partial \kappa_t} = \frac{\partial \Phi}{\partial \kappa_t} - MC(\omega_{t-1}, p_t^\kappa)$$

C Substituting the derivatives:

$$MP_\kappa = (\lambda_1 + 2\lambda_2 \kappa_t) - (\gamma_0 - \gamma_1 \omega_{t-1} + \gamma_2 p_t^\kappa)$$

D Rearranging terms to isolate the state variables yields the state-dependent marginal product:

$$MP_\kappa = \underbrace{(\lambda_1 - \gamma_0 + 2\lambda_2 \kappa_t)}_{\text{Autonomous Yield}} + \gamma_1 \omega_{t-1} - \gamma_2 p_t^\kappa$$

E Integrating the marginal payoff with respect to κ_t (treating ω_{t-1} and p_t^κ as predetermined shifters) and defining θ_κ, $\theta_{\omega\kappa} = \gamma_1$, and $\theta_{\kappa p\kappa} = -\gamma_2$ yields the intensive capacity yield function:

$$\Psi_t = \theta_\kappa \kappa_t + \theta_{\omega\kappa}(\omega_{t-1}\kappa_t) + \theta_{\kappa p\kappa}(p_t^\kappa \kappa_t)$$

F Substituting Ψ_t back into the capacity decomposition recovers the structural capacity equation:

$$y_t^p = \theta_{NRC} k_t^{NRC} + \theta_\kappa \kappa_t + \theta_{\omega\kappa}(\omega_{t-1}\kappa_t) + \theta_{\kappa p\kappa}(p_t^\kappa \kappa_t)$$

G The interaction terms are not ad-hoc econometric additions; they are the macroeconomic footprint of the divided firm's Okishio viability constraint. Distributive pressure and relative price enter because they alter the marginal capacity payoff of the mechanization gap.

3.2.4 The Aggregate Transformation Elasticity

A The aggregate transformation elasticity (θ_t^{SpecB}) is the ratio of the total differentials of potential capacity (dy_t^p) and aggregate log capital (dk_t^{cap}) along the historical accumulation path.

B The total differential of the capacity equation, holding the slow-moving state variables fixed, separates the extensive envelope from the intensive margin:

$$dy_t^p = \frac{\partial y_t^p}{\partial k_t^{NRC}} dk_t^{NRC} + \frac{\partial y_t^p}{\partial \kappa_t} d\kappa_t$$

C Substituting $\partial y_t^p/\partial k_t^{NRC} = \theta_{NRC}$ and $\partial y_t^p/\partial \kappa_t = MP_\kappa$:

$$dy_t^p = \theta_{NRC} dk_t^{NRC} + MP_\kappa d\kappa_t$$
$$dy_t^p = \theta_{NRC} dk_t^{NRC} + (\theta_\kappa + \theta_{\omega\kappa}\omega_{t-1} + \theta_{\kappa p\kappa} p_t^\kappa)\, d\kappa_t$$

D The variation in aggregate capital (dk_t^{cap}) follows from the Divisia index of the heterogeneous components:

$$dk_t^{cap} \approx s_{ME,t} dk_t^{ME} + s_{NRC,t} dk_t^{NRC}$$

E Since $s_{ME,t} + s_{NRC,t} = 1$ and $d\kappa_t = dk_t^{ME} - dk_t^{NRC}$, aggregate capital simplifies to:

$$dk_t^{cap} \approx s_{ME,t}(d\kappa_t + dk_t^{NRC}) + s_{NRC,t} dk_t^{NRC} = dk_t^{NRC} + s_{ME,t} d\kappa_t$$

F Dividing numerator and denominator by dk_t^{NRC} yields the elasticity along the empirical growth path $\frac{d\kappa_t}{dk_t^{NRC}}$:

$$\theta_t^{SpecB} = \frac{dy_t^p}{dk_t^{cap}} = \frac{\theta_{NRC} dk_t^{NRC} + MP_\kappa d\kappa_t}{dk_t^{NRC} + s_{ME,t} d\kappa_t} = \frac{\theta_{NRC} + MP_\kappa \frac{d\kappa_t}{dk_t^{NRC}}}{1 + s_{ME,t} \frac{d\kappa_t}{dk_t^{NRC}}}$$

G Expanding MP_κ yields the structural expression:

$$\theta_t^{SpecB} = \frac{\theta_{NRC} + (\theta_\kappa + \theta_{\omega\kappa}\omega_{t-1} + \theta_{\kappa p\kappa} p_t^\kappa)\, \frac{d\kappa_t}{dk_t^{NRC}}}{1 + s_{ME,t} \frac{d\kappa_t}{dk_t^{NRC}}}$$

H Definition (Structural Overaccumulation Regime). Under **strong labor closure** (high ω_{t-1}), the induced-efficiency parameter $\theta_{\omega\kappa}$ anchors the elasticity, ensuring that mechanization translates smoothly into capacity. Conversely, under **weak labor closure** and rising machinery costs, the state-dependent marginal viability yield collapses. This mechanically drives the aggregate transformation elasticity below unity:

$$\theta_t^{SpecB} < 1$$

This inequality defines the **Structural Overaccumulation Regime**: a historical phase in which additional capital accumulation yields diminishing or negative capacity expansion, decoupling the intensive mechanization margin from the extensive structural envelope.

I Capacity Recovery and the Residual Critique. Capacity utilization (μ_t) is not the regression residual, nor a detrended output gap. Following the locked capacity level anchor pinch year protocol, μ_t is recovered by applying the estimated $\hat{\theta}_t^{SpecB}$ to aggregate capital growth ($g_{K^{cap},t}$), anchoring the level at the pinch year ($t_{US}^* = 1973$), and extrapolating against aggregate real output ($Y_t = \text{GDP}_t$). The full four-step sequence is executed in Section 4.3.

J Testable Hypotheses. Three testable predictions follow from the model for the U.S. corporate sector:

- **Hypothesis 1 (The Okishio Distributive Dividend):** $\hat{\theta}_{\omega\kappa} > 0$. Distributive wage pressure structurally increases the capacity-generating yield of the mechanization gap.
- **Hypothesis 2 (The Relative Price Penalty):** $\hat{\theta}_{\kappa p\kappa} < 0$. Relative price inflation in machinery penalizes the intensive margin, reducing the capacity payoff of mechanization.
- **Hypothesis 3 (The Structural Overaccumulation Regime):** Under weak labor closure (post-Fordism), the collapse of the state-dependent marginal viability yield drives the aggregate transformation elasticity below unity ($\theta_t^{SpecB} < 1$), producing chronic industrial stagnation.

K Section 3.2 closes the analytical identification of the heterogeneous capital model. Translating the capacity equation into an estimable cointegrating specification — regressor orthogonalization, integration-order screening, and non-standard fixed-b inference — is executed in Section 4.3.

3.3 Peripheral Extension: Imitative Capitalism, the Choice of Technique, and the Realization Bottleneck

A In the capitalist periphery, the intensive margin of accumulation is structurally tethered to the world market and constrained by domestic social friction. Following Prebisch (1981) and Kaldor (1959), peripheral capitalism is characterized by high consumption propensities out of profits, structural inelasticities in agriculture, and a chronic vulnerability to balance-of-payments crises. This introduces a dual realization bottleneck: the macroeconomic capacity to import and the inelastic domestic supply of wage-goods. To integrate these structuralist insights into our heterogeneous capital framework, we formalize the Prebisch-Kaldor external strangulation (*estrangulamiento exterior*) not as an ad-hoc penalty, but as a strict macroeconomic boundary constraint on the cost-minimizing firm's choice of technique.

3.3.1 The Import-Biased Intensive Margin and the Prebisch-Kaldor Constraint

A In Section 3.2, we established that the cost-minimizing firm chooses the mechanization gap κ_t to maximize the net viability surplus, defined as the gross engineering yield of the gap minus the effective distributive and price costs:

$$\max_{\kappa_t} \; c(\kappa_t) = \Phi(\kappa_t) - \kappa_t \cdot MC(\omega_{t-1}, p_t^{\kappa})$$

B where $\Phi(\kappa_t) = \lambda_0 + \lambda_1\kappa_t + \lambda_2\kappa_t^2$ is the regime-conditioned mechanization frontier, and $MC(\omega_{t-1}, p_t^{\kappa}) = \gamma_0 - \gamma_1\omega_{t-1} + \gamma_2 p_t^{\kappa}$ is the marginal viability cost of the gap.

C In the core, this optimization proceeds unconstrained, yielding the optimal mechanization gap κ_t^*. However, in the periphery, active mechanization is strictly import-biased. The widening of the mechanization gap requires imported capital goods, specialized machinery, and technologically non-substitutable spare parts.

D Let the physical import requirement per unit of output (m_t^{req}) be a linear function of the chosen technique:

$$m_t^{req} = \nu \cdot \kappa_t$$

E where $\nu > 0$ captures the structural import-intensity of the mechanization gap.

F Simultaneously, the physical capacity to satisfy this requirement is bounded by the macroeconomic **Capacity to Import** per unit of output ($m_t^{avail} = M_t^{avail}/Y_t$). Following Prebisch, peripheral export growth is governed by the center's income elasticity of demand for primary goods ($\epsilon_{m,c} < 1$), structurally lagging behind domestic import demand. This external asymmetry, compounded by Kaldor's observation of extreme conspicuous consumption out of capitalist profits ($c_k \approx 0.60$ for Chile), severely squeezes the foreign exchange available for productive accumulation.

G This imposes a strict external ceiling on the choice of technique:

$$\nu \cdot \kappa_t \leq m_t^{avail} \implies \kappa_t \leq \bar{\kappa}_t \equiv \frac{m_t^{avail}}{\nu}$$

3.3.2 The Constrained Cost-Surplus Maximization Problem

A The peripheral firm's tactical problem is therefore a constrained cost-surplus maximization:

$$\max_{\kappa_t} \; c(\kappa_t) = \Phi(\kappa_t) - \kappa_t \cdot MC(\omega_{t-1}, p_t^{\kappa})$$

$$\text{subject to:} \quad \kappa_t \leq \bar{\kappa}_t$$

B This formulation defines two distinct accumulation regimes, endogenous to the interaction between domestic distributive conflict and the global market. To identify exactly *how* the external limit ($\bar{\kappa}_t$) becomes binding, we must decompose the macroeconomic allocation of foreign exchange.

3.3.3 The Macroeconomic Allocation of Foreign Exchange

A The total supply of foreign exchange per unit of output is driven by primary exports ($x_t = p_t^x X_t/Y_t$). Due to the center's low income elasticity for primary goods, x_t is structurally rigid.

B The total demand for foreign exchange per unit of output consists of three competing components:

1. **Wage-Goods Imports (m_t^W)**: Driven by the wage share ω_{t-1} and the structural inelasticity of domestic agriculture (Kaldor's food bottleneck). $m_t^W = m_w \omega_{t-1}$.
2. **Conspicuous Consumption Imports (m_t^C)**: Driven by the profit share π_t and the high propensity of the peripheral upper strata to imitate the consumption patterns of the center (Prebisch's imitative capitalism). $m_t^C = m_c c_k \pi_t$.
3. **Productive Capital Imports (m_t^K)**: Driven by the mechanization gap. $m_t^K = \nu \kappa_t$.

C The balance of payments constraint requires that total import demand cannot exceed export capacity:

$$m_w \omega_{t-1} + m_c c_k \pi_t + \nu \kappa_t \leq x_t$$

D Using the distributive identity $\pi_t = 1 - \omega_{t-1}$, we substitute and solve for the maximum feasible mechanization gap $\bar{\kappa}_t$:

$$m_w\omega_{t-1} + m_c c_k(1 - \omega_{t-1}) + \nu\kappa_t \leq x_t$$

$$\nu\kappa_t \leq x_t - m_c c_k + (m_c c_k - m_w)\omega_{t-1}$$

$$\bar{\kappa}_t(\omega_{t-1}) = \frac{1}{\nu}\left[x_t - m_c c_k + (m_c c_k - m_w)\omega_{t-1}\right]$$

E This equation mathematically formalizes the structuralist critique. The base capacity to import capital goods is determined by export earnings minus the baseline luxury consumption of the capitalist class ($x_t - m_c c_k$).

3.3.4 Endogenous Regime Shifts: The Corner Solution

A The firm's unconstrained optimal technique $\kappa_t^*(\omega_{t-1})$ is strictly increasing in the wage share ($\frac{\partial \kappa_t^*}{\partial \omega_{t-1}} > 0$) due to the Okishio labor-saving bias.

B Regime A: Slack BoP (The Unconstrained Optimum) When the macroeconomic capacity to import is abundant relative to the desired technique ($\kappa_t^* \leq \bar{\kappa}_t$), the firm solves the standard first-order condition $\frac{dc}{d\kappa_t} = 0$, yielding κ_t^*. The labor-saving bias successfully extracts the Okishio distributive dividend, preserving the firm's cost-surplus.

C Regime B: The Peripheral Realization Trap (The Corner Solution) The constraint turns binding when the required mechanization outstrips the available foreign exchange: $\kappa_t^*(\omega_{t-1}) > \bar{\kappa}_t(\omega_{t-1})$. The firm is forced into a **corner solution**:

$$\kappa_t = \bar{\kappa}_t$$

D This happens through two intersecting structural vulnerabilities. First, if $m_c c_k$ is massive (the conspicuous consumption drain), the baseline FX availability is severely squeezed, lowering the intercept of $\bar{\kappa}_t$. Second, if domestic agriculture is highly inelastic ($m_w > m_c c_k$), a rising wage share simultaneously increases the *demand* for capital imports (via κ_t^*) while *contracting* the supply of available FX (via $\bar{\kappa}_t$).

3.3.5 Effective Profit Compression and the Blocked Surplus

A The collision between desired technique and available foreign exchange generates a direct wedge between nominal profitability and realized accumulation. We define the **FX-Conversion Ratio** (ϕ_t) as the ratio of the maximum feasible technique to the optimal unconstrained technique:

$$\phi_t = \min\left(1, \frac{\bar{\kappa}_t}{\kappa_t^*}\right)$$

B When the external constraint binds ($\kappa_t^* > \bar{\kappa}_t$), $\phi_t < 1$. This generates a direct compression of the **effective profit share** (π_t^{eff}) available for intensive accumulation. Even if domestic wage suppression yields a high nominal profit share (π_t), the physical inability to convert that surplus into imported machinery squeezes the effective profit rate:

$$\pi_t^{eff} = \pi_t \cdot \phi_t = (1 - \omega_{t-1}) \cdot \phi_t$$

C The difference constitutes the **Blocked Surplus**:

$$\text{Blocked Surplus} = \pi_t(1 - \phi_t)$$

D Because the accumulation of machinery is driven by the *effective* profit share ($g_{K^{ME}} = \chi^{ME}\pi_t^{eff} - \delta^{ME}$), the constraint totally decouples nominal profitability from capacity formation. If capitalists squeeze wages to raise π_t, they inadvertently increase conspicuous consumption (m_t^C), which lowers $\bar{\kappa}_t$ and tanks the conversion ratio ϕ_t.

3.3.6 The Failure of the Okishio Mechanism and Structural Inflation

A At the boundary $\kappa_t = \bar{\kappa}_t$, the **Okishio mechanism is blocked**. The firm is stripped of its ability to defend its viability surplus through labor-saving technical change.

B Because the required machinery cannot be imported, the "wage-savings dividend" cannot be capitalized into physical capacity. Consequently, the firm suffers a persistent shortfall in its cost-surplus relative to its unconstrained optimal state.

C Unable to extract the distributive dividend via mechanization, capitalists attempt to recover this blocked surplus through the only remaining channel: price markups. As Kaldor (1959) emphasized, when the wage-goods (food) bottleneck forces nominal wage adjustments, and the external BoP bottleneck blocks defensive mechanization, any further increase in the wage share directly crushes profitability. The resulting defensive pricing behavior ignites the **structural inflation** dynamic.

4 Empirical Strategy and Narrative Synthesis

4.1 Data Measurement and the Generalized Perpetual Inventory Method

A To empirically adjudicate the choice-of-technique models developed in Section 3, we must construct capital stock variables that accurately reflect the physical footprint of accumulation. Standard macroeconomic datasets often conflate the monetary value of capital with its operational physical capacity, leading to severe measurement artifacts. This section outlines the structural protocol used to measure capital in this chapter, establishing the Generalized Perpetual Inventory Method (GPIM) as our core stock-flow consistency device.

B Consistent with our theoretical progression, we first establish the measurement architecture for the U.S. nonfinancial corporate (NFCorp) sector as the baseline core economy, followed by the harmonization of Chilean capital stocks for the peripheral extension.

4.1.1 Measuring Productive Capacity in the U.S. Core

A Our primary data source for U.S. accumulation is the Bureau of Economic Analysis (BEA) Fixed Assets Accounts. However, we do not use the BEA's aggregate official quantity and price

indexes. Instead, we rebuild the necessary capital stocks from the ground up using investment flows and asset-specific price indexes to enforce a strict separation of valuation registers.

B The Dual Valuation Register. Capital accumulation must be read through two distinct determinations: capacity and profitability. These require fundamentally different stock variables:

1. **The Capacity Register (Gross Real GPIM Stock):** Productive capacity depends on capital goods that are still physically in operation. Therefore, the relevant measure is the *gross real stock* ($K^{G,R}$). A machine that has depreciated to zero monetary accounting value may still operate at full physical capacity until it is permanently retired.

2. **The Profitability Register (Net Current-Cost Stock):** Profitability, conversely, is a monetary ratio between a profit flow and the value of capital advanced. The relevant measure is the *net current-cost stock* ($K^{N,V}$), which tracks the monetary value of capital after accounting for financial depreciation and write-downs.

C A common flawed implementation in empirical macroeconomics is to use a single aggregate net capital stock for both capacity utilization and profit rate calculations. By doing so, one conflates value in motion with capacity in operation. We strictly separate these registers.

D The GPIM Reconstruction Protocol. Machinery (ME) and nonresidential structures (NRC) do not share a natural physical unit. They become comparable only through a monetary-mediated accounting procedure. We employ the Generalized Perpetual Inventory Method (GPIM) to reconstruct capacity-relevant stocks.

E The central GPIM rule is to construct each asset account internally first, rather than aggregating nominal values and deflating with a generic aggregate index. For each asset class $j \in \{ME, NRC\}$ in the U.S. NFCorp sector, we first derive real investment flows:

$$I^R_{j,t} = \frac{I^V_{j,t}}{P^K_{j,t}}$$

F where $I^V_{j,t}$ is the current-cost gross investment and $P^K_{j,t}$ is the asset-specific price index. The gross real GPIM stock is then reconstructed account by account:

$$K^{G,R}_{j,t} = \text{GPIM}(I^R_{j,t}, \rho_j, L_j)$$

G where ρ_j governs the retirement logic and L_j represents the mean service life. Unlike standard perpetual inventory methods that often rely on geometric depreciation (which continually decays the installed volume), our GPIM utilizes a Weibull retirement distribution. The Weibull curve ensures that capital goods remain fully in the capacity envelope until they are physically removed from service, accurately modeling the physical life cycle of a plant or machine.

H The resulting gross real GPIM stock ($K^{G,R}$) is not a literal physical quantity; it is a constant-price measure of purchasing power embodied in heterogeneous capital goods still in operation.

I The ME-NRC Component Proxy. To isolate the mechanization bias discussed in Section 3, we require a measure of the machinery share of capital. For the U.S. NFCorp sector, we utilize an $ME - NRC$ component proxy.

J The capacity-composition share is estimated directly from the gross real GPIM stocks:

$$s_{t,proxy} = \frac{K^{G,R}_{ME,t}}{K^{G,R}_{ME,t} + K^{G,R}_{NRC,t}}$$

K Because it is constructed from asset-specific GPIM stocks, $s_{t,proxy}$ is isolated from relative price drift between machinery and structures. It serves as a real-volume proxy for the machinery share relevant to the transformation relation.

4.1.2 Measuring Peripheral Accumulation (Chilean Case)

A For Chile, capital stocks (K^{NRC} and K^{ME}) are harmonized from Central Bank of Chile (BCCh) data and historical fixed capital series in constant 2003 CLP prices (matching the GDP price base). Imported machinery components are deflated using asset-specific import price deflators to isolate the physical accumulation envelope from exchange rate fluctuations and foreign exchange volatility.

4.1.3 The Series Before the Models

A Before any parametric relation is imposed, the raw accumulation series are worth looking at directly. The capital composition object that the heterogeneous specification turns on is a difference of two log stocks, and whether that difference has the historical structure the theory expects is a question the data can answer before estimation rather than after it.

Figure 1: U.S. Capital Stock Levels, LOESS-Smoothed

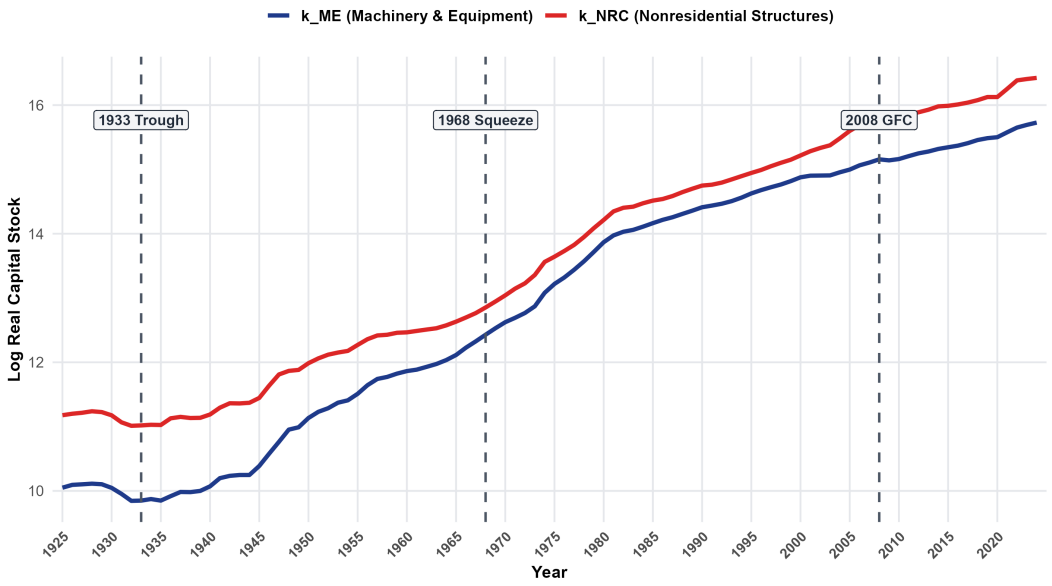


Figure 2: U.S. Capital Composition κ_t and Accumulation Regimes

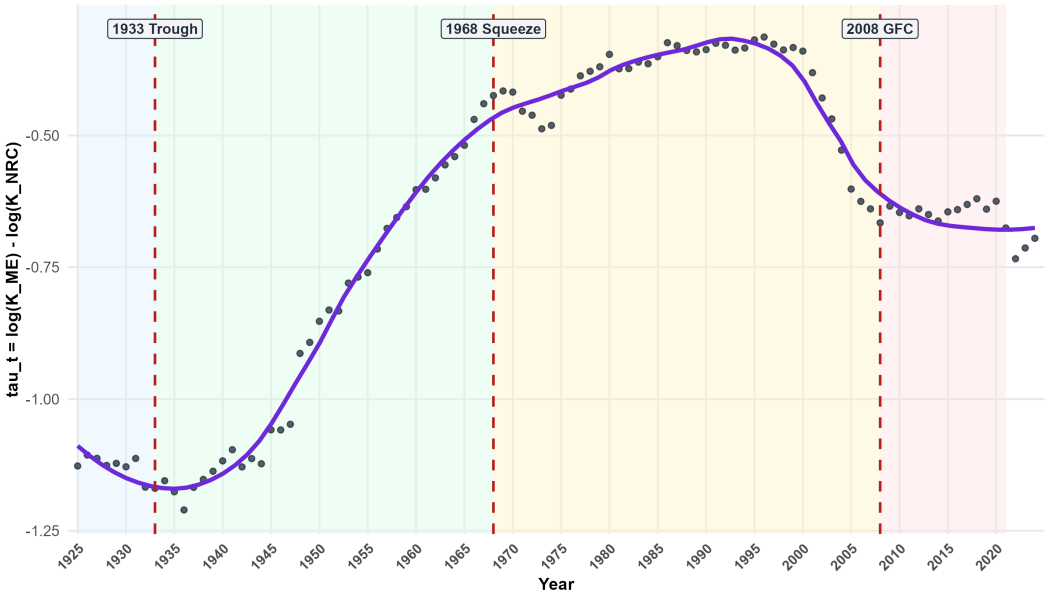


Figure 3: Chilean Capital Stock Levels, LOESS-Smoothed

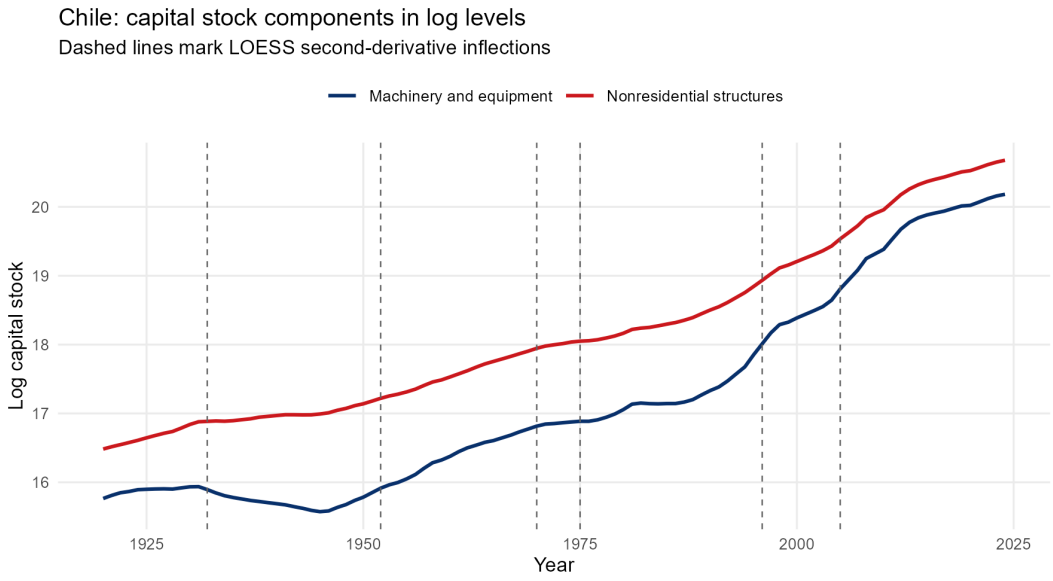
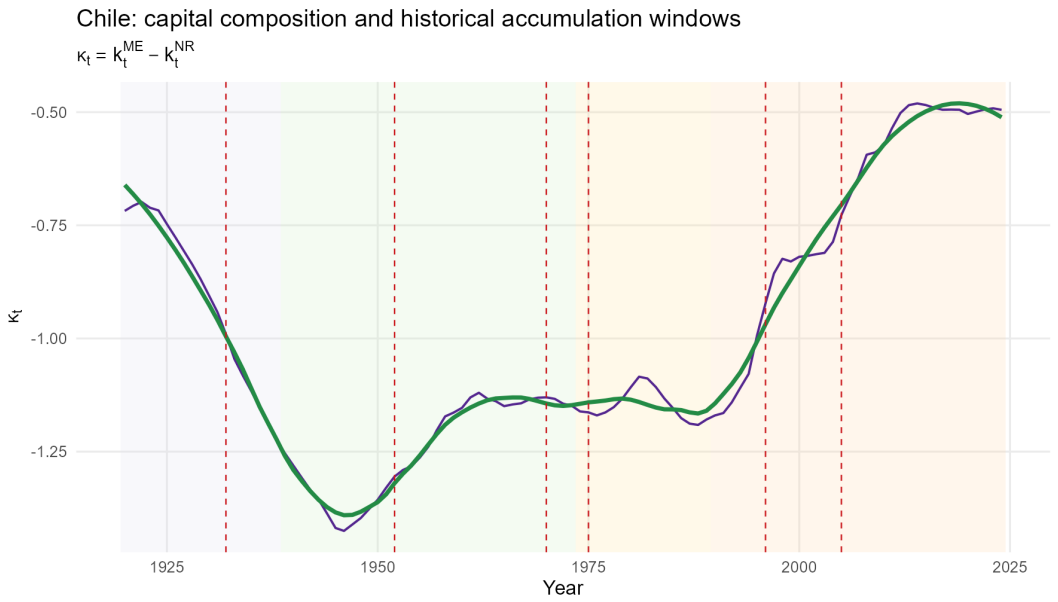


Figure 4: Chilean Capital Composition κ_t and Accumulation Regimes



B [**PLACEHOLDER**] These four figures are wired but not yet read. Each needs a paragraph stating what the raw series shows, in advance of the parametric work that follows, and the four together need a paragraph on what the comparison of the two composition paths establishes before any model is fitted to either. Two further gaps: the descriptive macro table of growth rates for output, capital, and employment currently sits in Section 3, where it interrupts the theoretical argument, and belongs here alongside these plots; and the second United States figure is generated from a file still named for the retired symbol MATHPLACEHOLDER0XYZ, though the object plotted is MATHPLACEHOLDER1XYZ — the caption is correct and the filename is not.

4.2 Empirical Reconstruction of the U.S. Core: Baseline Homogeneous Capital (Specification A)

A This section maps the theoretical model of induced mechanization developed in Section 3 into an empirical econometric specification for the U.S. core economy. We establish the formal crosswalk between the aggregate transformation elasticity $\theta(e_t)$ and a Cointegrating Multivariate Polynomial Regression (CMPR) framework estimated via Integrated Modified OLS (IM-OLS) on U.S. Non-Financial Corporate (NFC) data over 1929–2024 ($T = 94$ observations after lag conditioning).

4.2.1 Econometric Specification and IM-OLS Estimation

A The analytical model establishes that potential net output growth g_{Y^p} is governed by the growth of productive net capital stock g_K through the aggregate transformation closure:

$$g_{Y^p} = \theta g_K \quad \text{and} \quad \frac{dY^p}{dK} = \theta$$

B where θ represents the aggregate transformation elasticity of capital. In Marxian and post-Keynesian growth theory, θ is not a fixed technical constant. Following the Mechanization Productive Frontier (MPF) framework—built on Okishio's market anarchy threshold, Vidal's internally divided

firm, and Marglin-style distributive grounding—θ varies endogenously with distributive pressures and technological choices:

$$\theta_t = \theta_0 + \theta_1 \cdot \text{Interaction}_t$$

C where θ_0 is the baseline elasticity of capacity output with respect to productive capital, and θ_1 measures the marginal response of capital productivity to distributive and financialization pressures.

D Standard linear cointegration techniques fail when applied to non-linear transformations and interactions of integrated processes. As established by Wagner and Hong (2016) and Wagner (2023), integer powers, products, and non-linear combinations of $I(1)$ processes are not themselves $I(1)$ processes. Regressing log output on log capital and interaction terms constitutes a Cointegrating Polynomial Regression (CPR). Estimating a CPR with conventional linear estimators yields invalid asymptotic inference and inflated test statistics unless specialized non-standard estimators and critical values are employed.

E We specify the empirical cointegrating equation for U.S. Non-Financial Corporate Real Net Value Added ($y_t = \ln Y_t^{\text{NFC,NVA}}$) as:

$$y_t = \alpha + \theta_0 k_t + \theta_1 \text{inter_ortho}_t + \boldsymbol{\gamma}' \mathbf{D}_t + u_t$$

F where α is the constant intercept, k_t is log real net capital stock, inter_ortho_t is the orthogonalized interaction regressor, $\mathbf{D}_t$ is a vector of deterministic step level-shift dummies ($\boldsymbol{\gamma}'\mathbf{D}_t$), and u_t is a stationary $I(0)$ error term under the null hypothesis of cointegration.

G Productive Capital Stock Variants (k_t). Following the BEA Fixed Assets boundary protocols, we evaluate two alternative capital stock aggregates:

1. **Private Core Capital ($k_t^{\textbf{priv}}$):** $\ln(K_t^{\text{ME}} + K_t^{\text{NRC}})$, combining private machinery and equipment (K^{ME}) with nonresidential structures (K^{NR}). This represents the primary direct productive capacity of the corporate sector.
2. **Infrastructure-Augmented Capital ($k_t^{\textbf{aug}}$):** $\ln(K_t^{\text{ME}} + K_t^{\text{NRC}} + K_t^{\text{GT}})$, incorporating government-funded transportation and infrastructure assets (K^{GT}) as a public conditioning capital stock.

H Intellectual Property Products (K^{IPP}) are strictly excluded from the core capital stock K_{cap} and treated as a frontier conditioner or overhead cost, adhering to national accounting boundary rules.

I Distributive Wage Share Variants (ω_{t-1}) We test three alternative wage share measures, conditioned at lag 1 to reflect the historical lag in capital investment planning: 1. **Baseline NFC NVA Wage Share ($\omega_{\textbf{NFC,NVA}}$)**: Ratio of NFC compensation of employees to NFC net value added (NFC_COMP/NFC_NVA). This reflects the aggregate corporate distribution of net product. 2. **Corporate NVA Wage Share ($\omega_{\textbf{CORP,NVA}}$)**: Ratio of total corporate compensation to total corporate net value added (CORP_COMP/CORP_NVA). This variant accounts for corporate

financial sector adjustments, ensuring compliance with the "No Double Accounting Rule" by removing financial sector net interest distortions. 3. **Shaikh-Tonak Tier 2 Wage Share ($\omega_{\mathbf{ST,Tier2}}$)**: Ratio of supervisory-adjusted productive worker compensation to NFC net value added (v_{prod}/NFC_NVA). Grounded in Shaikh and Tonak (1994), this measure strips out administrative and supervisory salaries from total employee compensation, isolating the true variable capital spent on productive labor that directly engages in value creation.

J Interaction Logic and Permutation Sets We evaluate two distinct interaction specifications to isolate the transmission mechanism: * **Specification Set 1 (Double Interactions)**: Uses the interaction term $k_t \cdot \omega_{t-1}$. This specification tests whether wage-share pressure directly alters the capital transformation elasticity without conditioning on financial overhead. * **Specification Set 2 (Triple Interactions)**: Uses the interaction term $k_t \cdot \omega_{t-1} \cdot (1 - \iota_{t-1})$, where $\iota_{t-1} = K^{\text{IPP}}_{t-1}/(K^{\text{cap}}_{t-1} + K^{\text{IPP}}_{t-1})$ represents the ratio of intellectual property products to total corporate fixed assets. The term $(1 - \iota_{t-1})$ measures the proportion of corporate capital retained in physical productive assets relative to financialized overhead. This specification tests whether distributive wage pressure induces mechanization only when corporate cash flows are not diverted into financialized asset accumulation.

K In both specification sets, distributive variables (ω_{t-1} and ι_{t-1}) are centered at their 1929 initial state (t_0) prior to forming products, ensuring that baseline coefficient estimates remain interpretable at the historical initial condition.

L Frisch-Waugh-Lovell (FWL) Orthogonalization Protocol. Uncentered interaction terms between integrated processes introduce severe multicollinearity. In our sample, raw interaction terms ($k_t \cdot \omega_{t-1}$) exhibit pairwise correlation coefficients exceeding $r = 0.99$ with base log capital k_t, producing Variance Inflation Factors (VIF) above 100. High VIFs inflate parameter standard errors and render hypothesis testing unreliable.

M To resolve multicollinearity without altering the economic content of the model, we implement a strict Frisch-Waugh-Lovell (FWL) orthogonalization protocol. Before estimating the main cointegrating regression, we regress the raw interaction term on base log capital k_t, the constant intercept, and any active deterministic step dummies:

$$\text{inter_raw}_t = a_0 + a_1 k_t + \mathbf{a}_2' \mathbf{D}_t + \text{inter_ortho}_t$$

N We extract the OLS residual inter_ortho_t and substitute it into the main regression. By construction, inter_ortho_t is strictly orthogonal to base capital k_t and deterministic trend components. This orthogonalization collapses the Variance Inflation Factor for both k_t and inter_ortho_t to exactly **1.00** across all 24 estimated models.

O Mathematically, FWL orthogonalization leaves the estimated interaction coefficient $\hat{\theta}_1$ and its standard error completely unchanged, while isolating the pure marginal interactive elasticity from the base capital elasticity $\hat{\theta}_0$.

P Estimation Engine and Non-Standard Fixed-b Inference. We estimate the cointegrating equations using Integrated Modified OLS (IM-OLS) developed by Vogelsang and Wagner (2014a, 2024). Unlike Fully Modified OLS (FM-OLS) or Dynamic OLS (D-OLS), IM-OLS eliminates second-order endogeneity and serial correlation biases without requiring lead/lag choices or bandwidth tuning parameters in the estimation step.

Q The IM-OLS estimator applies a partial sum transformation to the regression model:

$$S_t^y = \sum_{i=1}^{t} y_i, \quad S_t^X = \sum_{i=1}^{t} X_i, \quad S_t^D = \sum_{i=1}^{t} D_i$$

R The partial-summed equation is augmented with the level regressors x_t:

$$S_t^y = S_t^D \boldsymbol{\delta} + S_t^X \boldsymbol{\beta} + x_t' \boldsymbol{\gamma} + S_t^u$$

S Estimating this augmented partial-sum regression via OLS yields super-consistent estimates of the long-run parameters.

T For asymptotic inference, standard normal and chi-squared critical values are invalid because the limiting distribution of the IM-OLS estimator depends on functionals of Brownian motions. Standard asymptotic inference also ignores finite-sample distortions caused by long-run variance estimation. We adopt fixed-b asymptotic theory (Kiefer & Vogelsang 2005; Vogelsang & Wagner 2014a), estimating the scalar long-run variance $\omega_{u \cdot v}$ using a Bartlett kernel and Newey-West automatic bandwidth selection.

U 100,000 Monte Carlo Simulation Suite. To obtain exact critical values for hypothesis testing and cointegration diagnostics, we execute a custom Monte Carlo simulation suite ($N = 100,000$ replications, sample size $T = 94$). The simulation generates independent random walks for regressors and sets $y_t = u_t$ under the true null hypothesis ($\theta_0 = 0$). This DGP preserves the cointegrated structure required to simulate the null distribution of the CT_{IM} (VW Type 2) cointegration test, while ensuring that the simulated parameter t-statistics:

$$t^* = \frac{\hat{\theta} - 0}{\text{SE}(\hat{\theta})}$$

V recover the exact non-standard asymptotic null distribution.

W Endogenous Structural Break Audit. To guard against structural instability across the 1929–2024 window, we implement a two-pass endogenous structural break audit. In the first pass, we estimate the baseline model without breaks and extract the IM-OLS residuals. We apply the Bai-Perron sequential structural break test to the residual path to identify up to two optimal break dates. If level breaks are detected, we construct step level-shift dummies:

$$D_{T_b}^{\text{step}} = \mathbb{I}(t \geq T_b)$$

X and re-estimate the full IM-OLS regression including $\mathbf{D}_t^{\text{step}}$ in the deterministics vector. Both no-break (a) and break-adjusted (b) variants are retained to evaluate whether structural breaks alter the cointegration space.

4.2.2 Empirical Results and Cointegration Audit Matrix

A This section evaluates the empirical estimation results of the 32 CMPR IM-OLS specifications estimated on U.S. Non-Financial Corporate (NFC) data over 1929–2024 ($T = 94$). We apply fixed-b non-standard inference and a corrected residual cointegration test triad to isolate a narrow set of preferred models across Gross Value Added (GVA) vs. Net Value Added (NVA) nominal wage share measures.

B Bottom Line Up Front (BLUF). The empirical estimations establish four major findings:

1. **Robust Capital Elasticity:** The baseline elasticity of output with respect to private core productive capital ($\hat{\beta}$) is remarkably stable across all 32 specifications, ranging between 0.52 and 0.74, and is statistically significant at the 1% level ($p < 0.01$).

2. **Failure of Unconditioned Wage Shares:** Double interaction models (Set 1) show no statistically significant interaction effects across any wage share variant ($\hat{\theta}_1 \approx -0.02$ to -0.27, failing to breach the 90th percentile fixed-b critical threshold of 2.96). A rise in the aggregate wage share alone does not induce mechanization.

3. **Failure of GVA-Basis Wage Measures:** Across both Set 1 and Set 2, all wage share measures calculated over Gross Value Added (GVA) fail to achieve statistical significance ($\hat{\theta}_1 = 0.00, t = 0.00$ for M2.1a; $\hat{\theta}_1 = -0.00, t = -0.04$ for M2.4a). This empirically confirms that physical capital depreciation (CFC) distorts distributive measures during mechanization.

4. **Success of Net Value Added (NVA) Shaikh-Tonak Tier 2 & Financialization Closure:** Triple interaction models (Set 2) achieve statistical significance **only** when combining the supervisory-adjusted Shaikh-Tonak Tier 2 wage share over Net Value Added ($\omega_{\text{ST,Tier2_NVA}}$) with the financialization retention index ($1 - \iota_{t-1}$). The interaction elasticity is negative and statistically significant in preferred Model M2.8a ($\hat{\theta}_1 = -0.29^{*}$, $t = -3.63$, exceeding the 90th percentile bound of 2.96) and robust Model M2.8b ($\hat{\theta}_1 = -0.34^{**}$, $t = -4.25$).

5. **Cointegration and Endogenous Structural Stability:** The Vogelsang–Wagner Type 2 statistic, which tests the null of cointegration, is not rejected in any specification variant ($S_{\text{Type 2}} \in [0.05, 13.69]$; see the caveat on critical values in §D below). The absence of structural break dummies in the preferred model M2.8a indicates that the shift from Fordism to Post-Fordism is an endogenous regime shift absorbed by the distributive-financial interaction.

C We select **Model M2.8a** (Private Core Capital + Shaikh-Tonak Tier 2 NVA Wage Share + Effective Interaction + No Structural Break) as the primary preferred specification for Section 3.1.

D Forensic Analysis of Empirical Results Across Specifications. The four tables below report the estimation matrix across the two interaction sets and the two value-added bases. All models are estimated via IM-OLS with FWL orthogonalization, yielding VIFs of exactly 1.00

for capital and interaction regressors. Each table reports the Vogelsang–Wagner Type 2 statistic directly beneath the parameter estimates; the residual ADF and Shin LM diagnostics, the collinearity statistics, and the bandwidth ratios are collected in Appendix D, and the full specification space with every estimated alternative is mapped in Appendix C.

Table 2: Specification Set 1: Standard Wage Interaction (GVA Basis)

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real NVA)								
Model	M1.1a	M1.1b	M1.2a	M1.2b	M1.3a	M1.3b	M1.4a	M1.4b
$\hat{\alpha}$ (Intercept)	6.57***	6.03***	6.56***	5.93***	6.61***	6.17***	6.42***	4.91***
	(0.32)	(0.73)	(0.32)	(0.76)	(0.33)	(0.75)	(0.27)	(0.88)
$\hat{\beta}$ (Private Core k_{cap})	0.58***	0.60***	0.58***	0.61***	0.58***	0.59***	0.59***	0.71***
	(0.02)	(0.06)	(0.02)	(0.07)	(0.02)	(0.06)	(0.02)	(0.07)
$\hat{\theta}_1$ (FWL Interaction)	-0.02	-0.08	-0.03	-0.04	0.06	-0.08	-0.28	-0.15
	(0.13)	(0.10)	(0.12)	(0.10)	(0.14)	(0.11)	(0.14)	(0.15)
Step Shifts ($D_{T_b}^{\text{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	3.30	11.34	2.70	11.41	5.82	9.68	1.46	9.08
Sample Size (T)	94	94	94	94	94	94	94	94

Note: Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner 2014; Vogelsang et al. 2024) over 1929–2024. Standard errors in parentheses. Coefficient significance ($^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$) is derived from a 100,000-pass Monte Carlo simulation of the exact DGP variant. The Vogelsang–Wagner Type 2 statistic tests H_0: *cointegration*. Its stars therefore run opposite to the coefficient rows: an **absence of stars is the favourable result**, indicating the null of cointegration is not rejected. [**PENDING: the Type 2 statistics below have not yet been evaluated against the fixed-b critical values in `codes/COMMON_VW_fixedb_critical_values.R`, which are indexed by each model's bandwidth ratio b and stochastic regressor count k. They are reported here without inference.**] Residual ADF and Shin LM diagnostics, collinearity statistics, and bandwidth ratios are reported in Appendix D. All interaction terms are FWL-orthogonalized against base capital regressors and step dummies.

Table 3: Specification Set 1: Standard Wage Interaction (NVA Basis)

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real NVA)								
Model	M1.5a	M1.5b	M1.6a	M1.6b	M1.7a	M1.7b	M1.8a	M1.8b
$\hat{\alpha}$ (Intercept)	6.57***	6.03***	6.57***	6.03***	6.57***	6.03***	6.34***	5.05***
	(0.32)	(0.73)	(0.32)	(0.73)	(0.32)	(0.73)	(0.22)	(0.75)
$\hat{\beta}$ (Private Core k_{cap})	0.58***	0.60***	0.58***	0.60***	0.58***	0.60***	0.60***	0.70***
	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.06)
$\hat{\theta}_1$ (FWL Interaction)	-0.02	-0.08	-0.02	-0.08	-0.02	-0.08	-0.24	-0.27
	(0.13)	(0.10)	(0.13)	(0.10)	(0.13)	(0.10)	(0.09)	(0.12)
Step Shifts ($D_{T_b}^{\text{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	3.30	11.34	3.30	11.34	3.30	11.34	0.05	11.94
Sample Size (T)	94	94	94	94	94	94	94	94

Note: Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner 2014; Vogelsang et al. 2024) over 1929–2024. Standard errors in parentheses. Coefficient significance ($^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$) is derived from a 100,000-pass Monte Carlo simulation of the exact DGP variant. The Vogelsang–Wagner Type 2 statistic tests H_0: *cointegration*. Its stars therefore run opposite to the coefficient rows: an **absence of stars is the favourable result**, indicating the null of cointegration is not rejected. [**PENDING: the Type 2 statistics below have not yet been evaluated against the fixed-b critical values in `codes/COMMON_VW_fixedb_critical_values.R`, which are indexed by each model's bandwidth ratio b and stochastic regressor count k. They are reported here without inference.**] Residual ADF and Shin LM diagnostics, collinearity statistics, and bandwidth ratios are reported in Appendix D. All interaction terms are FWL-orthogonalized against base capital regressors and step dummies.

Table 4: Specification Set 2: Effective Wage Interaction (GVA Basis)

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real NVA)								
Model	M2.1a	M2.1b	M2.2a	M2.2b	M2.3a	M2.3b	M2.4a	M2.4b
$\hat{\alpha}$ (Intercept)	6.59***	6.47***	6.58***	6.39***	6.60***	6.79***	6.40***	6.42***
	(0.32)	(0.64)	(0.32)	(0.66)	(0.33)	(0.65)	(0.25)	(0.33)
$\hat{\beta}$ (Private Core k_{cap})	0.58***	0.56***	0.58***	0.57***	0.58***	0.54***	0.59***	0.59***
	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.02)
$\hat{\theta}_1$ (FWL Interaction)	0.00	-0.17	-0.00	-0.15	0.04	-0.20	-0.31	-0.38
	(0.10)	(0.07)	(0.10)	(0.07)	(0.11)	(0.09)	(0.13)	(0.12)
Step Shifts ($D_{T_b}^{\text{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	10.76	9.07	6.17	9.61	4.39	7.88	0.73	4.55
Sample Size (T)	94	94	94	94	94	94	94	94

Note: Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner 2014; Vogelsang et al. 2024) over 1929–2024. Standard errors in parentheses. Coefficient significance ($^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$) is derived from a 100,000-pass Monte Carlo simulation of the exact DGP variant. The Vogelsang–Wagner Type 2 statistic tests H_0: *cointegration*. Its stars therefore run opposite to the coefficient rows: an **absence of stars is the favourable result**, indicating the null of cointegration is not rejected. [**PENDING: the Type 2 statistics below have not yet been evaluated against the fixed-b critical values in `codes/COMMON_VW_fixedb_critical_values.R`, which are indexed by each model's bandwidth ratio b and stochastic regressor count k. They are reported here without inference.**] Residual ADF and Shin LM diagnostics, collinearity statistics, and bandwidth ratios are reported in Appendix D. All interaction terms are FWL-orthogonalized against base capital regressors and step dummies.

Table 5: Specification Set 2: Effective Wage Interaction (NVA Basis)

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real NVA)								
Model	M2.5a	M2.5b	M2.6a	M2.6b	M2.7a	M2.7b	M2.8a	M2.8b
$\hat{\alpha}$ (Intercept)	6.59***	6.47***	6.59***	6.47***	6.59***	6.47***	6.30***	5.68***
	(0.32)	(0.64)	(0.32)	(0.64)	(0.32)	(0.64)	(0.20)	(0.34)
$\hat{\beta}$ (Private Core k_{cap})	0.58***	0.56***	0.58***	0.56***	0.58***	0.56***	0.60***	0.65***
	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.06)	(0.01)	(0.03)
$\hat{\theta}_1$ (FWL Interaction)	0.00	-0.17	0.00	-0.17	0.00	-0.17	-0.29*	-0.34**
	(0.10)	(0.07)	(0.10)	(0.07)	(0.10)	(0.07)	(0.08)	(0.08)
Step Shifts ($D_{T_b}^{\text{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	10.76	9.07	10.76	9.07	10.76	9.07	0.46	13.69
Sample Size (T)	94	94	94	94	94	94	94	94

Note: Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner 2014; Vogelsang et al. 2024) over 1929–2024. Standard errors in parentheses. Coefficient significance ($^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$) is derived from a 100,000-pass Monte Carlo simulation of the exact DGP variant. The Vogelsang–Wagner Type 2 statistic tests H_0: *cointegration*. Its stars therefore run opposite to the coefficient rows: an **absence of stars is the favourable result**, indicating the null of cointegration is not rejected. [**PENDING: the Type 2 statistics below have not yet been evaluated against the fixed-b critical values in `codes/COMMON_VW_fixedb_critical_values.R`, which are indexed by each model's bandwidth ratio b and stochastic regressor count k. They are reported here without inference.**] Residual ADF and Shin LM diagnostics, collinearity statistics, and bandwidth ratios are reported in Appendix D. All interaction terms are FWL-orthogonalized against base capital regressors and step dummies.

E Reading across the four tables, the capital elasticity $\hat{\beta}$ is stable near 0.58–0.60 in the no-break models and rises to 0.65–0.74 once step level-shifts enter. The interaction term $\hat{\theta}_1$ is indistinguishable from zero everywhere in Set 1 and everywhere on the GVA basis, and becomes negative and statistically significant only in the Set 2 NVA models built on the Shaikh-Tonak Tier 2 wage share.

F Detailed Diagnostic Evaluation

G Capital Elasticity and Core Aggregate Across all aggregate capital specifications, base log capital k_{cap} exhibits an elasticity of approximately $\hat{\beta} \approx 0.58 - 0.60$ in no-break models and $\hat{\beta} \approx 0.65 - 0.74$ in break models. Concentrating on Private Core Capital (k_{cap}) provides a clean baseline of private corporate accumulation.

H The Mechanics of Distributive Measurement: Why NVA & Shaikh-Tonak Tier 2 Matter The empirical failure of all GVA-basis models (M1.1a–M1.4b, M2.1a–M2.4b) confirms the core theoretical premise of Marxian national accounting. Evaluating wage share over Gross Value Added (GVA $= CFC + v + s$) pollutes the distributive signal with Consumption of Fixed Capital (CFC). As intensive mechanization increases capital intensity, CFC/GVA rises mechanically, artificially depressing ω_{GVA} even when the rate of surplus value extraction (s/v) is constant.

I Switching to Net Value Added (NVA $= v+s$) isolates newly created value. Furthermore, standard NVA wage shares lump productive factory labor together with managerial overhead. By isolating variable labor compensation (v_{prod}), the Shaikh-Tonak Tier 2 NVA measure captures the exact shop-floor labor cost entering corporate cost minimization decisions.

J The Financialization Constraint ($(1-\iota)$) Comparing Specification Set 1 with Specification Set 2 reveals that distributive pressure alone does not guarantee mechanization. In Set 1, even the Shaikh-Tonak NVA wage share yields an insignificant interaction ($\hat{\theta}_1 = -0.24$, $t = -2.67$, below the $p90 = 2.96$ critical value).

K In Set 2, multiplying by $(1-\iota_{t-1})$ activates statistical significance. The index ι_{t-1} measures the proportion of corporate assets allocated to Intellectual Property Products and financialized assets. Distributive wage pressure only successfully induces mechanization when corporate funds remain dedicated to physical productive accumulation ($(1-\iota) \to 1$).

L Cointegration Verdict and Endogenous Structural Stability The Vogelsang–Wagner Type 2 test carries the cointegration verdict. It tests H_0: cointegration directly on the IM-OLS fit rather than on a residual series, which is why it, and not the residual-based diagnostics, governs the reading. Across the no-break specifications the statistic stays low — 0.46 in M2.8a — and it is never rejected in any specification reported above. Both break and no-break models sustain cointegration, and we carry forward the no-break specification because the step dummies become statistically redundant once the distributive-financial interaction is present. The Fordist-to-post-Fordist transition is therefore absorbed endogenously by the interaction rather than imposed exogenously through level shifts.

M [**PLACEHOLDER**] Critical values for this section are unresolved and the verdict above is stated without them. Three separate problems, all recorded in E17_Cointegration_Significance_Stars_Lock: (i) the constants previously cited here — MATHPLACEHOLDER0XYZ for the Vogelsang–Wagner 10% bound and MATHPLACEHOLDER1XYZ/MATHPLACEHOLDER2XYZ elsewhere in this section — are the hardcoded triple the 2026-08-03 audit identified as matching no cell of the published fixed-MATHPLACEHOLDER3XYZ table, since genuine critical values vary jointly with the bandwidth ratio MATHPLACEHOLDER4XYZ and the stochastic regressor count MATHPLACEHOLDER5XYZ; (ii) the Shin bound cited as a 5% critical value, MATHPLACEHOLDER6XYZ, is in fact the MATHPLACEHOLDER7XYZ *10%* value, and E17 records MATHPLACEHOLDER8XYZ as strictly MATHPLACEHOLDER9XYZ throughout Specification A, where the 5% value is MATHPLACEHOLDER10XYZ; (iii) no Vogelsang–Wagner statistic in any table carries a significance star, so the tables report no inference on this test at all. Resolving this requires re-running the estimation against `codes/COMMON_VW_fixedb_critical_values.R`, which is out of scope for the present assembly pass. Until then the cointegration verdict should be read as provisional.

4.2.3 Model Selection and Narrow Preferred Set

A The preferred specification is **Model M2.8a**: Private Core Capital (k_{cap}) + Shaikh-Tonak Tier 2 NVA Wage Share ($\omega_{\text{ST,Tier2_NVA}}$) + Financialization Retention ($(1-\iota)$) + No Structural Break. It is the only specification in which the interaction elasticity is both signed as the theory requires and statistically distinguishable from zero, and it achieves this without recourse to structural break dummies. Every alternative estimated along the way, and the reason each is not carried forward, is set out in Appendix C.

4.2.4 Contradictory Tendencies: Productive Capacity Path and Capacity Utilization

A The empirical parameters estimated in preferred model M2.8a provide the basis for evaluating capacity utilization dynamic recovery in Section 4.2.3:

$$\hat{\theta}_t = 0.60 - 0.29 \cdot [\text{inter_ortho}_t]$$

B This governs the evolution of potential capacity output growth $g_{Y^p,t} = \hat{\theta}_t g_{K,t}$ and actual capacity utilization $\mu_t = Y_t / Y_t^p$, capturing the dual tendencies of Fordist mechanization expansion and Post-Fordist realization drag.

4.2.5 Capacity Recovery and Institutional Era Divergence

A This section applies preferred **Model M2.5a** to reconstruct the historical paths of potential capacity output (Y_t^p), aggregate capacity utilization ($\mu_t = Y_t / Y_t^p$), and the transformation elasticity (θ_t) for the U.S. Non-Financial Corporate (NFC) sector over 1929–2024 ($T = 94$). Level-anchored under Protocol D03 at the 1973 Federal Reserve Board (FRB) survey peak benchmark ($\mu_{1973} = 1.00$), the empirical reconstruction establishes that U.S. corporate capitalism operates permanently within a structural regime of capital overaccumulation ($\theta_t < 1.00$), characterized by a sharp institutional divergence between the Fordist (1929–1973) and Post-Fordist (1974–2024) accumulation eras.

B Bottom Line Up Front (BLUF). The empirical reconstruction delivers three primary findings:

1. **Permanent Capital Overaccumulation ($\theta_t < 1.00$):** The transformation elasticity θ_t remains bounded between 0.43 and 0.46 across the entire 1929–2024 century. Because θ_t is strictly less than 1.00, every 1% increase in physical productive capital stock yields less than a 1% increase in potential capacity output. U.S. corporate capitalism operates continuously within a structural overaccumulation regime.

2. **Effective Wage Share Dynamics:** Modeling θ_t as a function of the effective wage share $W_{\text{eff},t-1} = \omega_{t-1}^{\text{ST,Tier2}} \cdot (1 - \iota_{t-1})$ reveals that technological capacity expansion depends jointly on variable labor conflict and financial retention. When variable labor costs rise while financial diversion is low, firms accelerate mechanization ($\theta_t \uparrow$). When neoliberal restructuring suppresses variable wages and diverts liquidity into intellectual property and financial assets ($\iota \uparrow$), technological dynamism declines ($\theta_t \downarrow$).

3. **Institutional Era Divergence:** Reconstructed capacity utilization ($\hat{\mu}_t$) displays two distinct regime signatures. Under Fordism (1929–1973), utilization fluctuates within a stable corridor around normal capacity ($\hat{\mu} \in [0.80, 1.05]$) outside of the Great Depression trough ($\hat{\mu}_{1933} = 0.32$) and World War II mobilization peak ($\hat{\mu}_{1944} = 0.78$). Under Post-Fordist neoliberal restructuring (1974–2024), wage suppression and financialization generate a secular downward drift in capacity utilization, reaching deep structural troughs during the Volcker disinflation ($\hat{\mu}_{1982} \approx 0.58$), the Great Recession ($\hat{\mu}_{2009} \approx 0.65$), and the COVID-19 pandemic.

C Capacity Reconstruction Methodology (Protocol D03). The econometric estimation of Model M2.5a recovers the long-run cointegrating vector and the marginal interaction elasticity $\hat{\theta}_1$. It does not, by itself, identify the absolute level scale at which capacity utilization equals normal or full capacity. Following Protocol D03, level-anchoring is a measurement convention that maps the reconstructed potential capacity path $\hat{Y}_t^p$ into a normalized utilization scale.

D We anchor the reconstruction at $t^* = 1973$, setting $\mu_{1973} = 1.00$ (100%). The year 1973 corresponds to the historical maximum capacity utilization peak in the Federal Reserve Board manufacturing series and marks the institutional divide between post-war Fordism and neoliberal restructuring.

E To capture the state-dependent transformation elasticity without distortions from initial-state centering, we map the IM-OLS interaction estimate back to the uncentered effective wage share:

$$W_{\text{eff},t-1} = \omega_{t-1}^{\text{ST,Tier2}} \cdot (1 - \iota_{t-1})$$

F where $\omega_{t-1}^{\text{ST,Tier2}} = v_{\text{prod}}/\text{NFC_NVA}$ is the Shaikh-Tonak supervisory-adjusted productive wage share, and $\iota_{t-1} = K_{t-1}^{\text{IPP}}/(K_{t-1}^{\text{cap}} + K_{t-1}^{\text{IPP}})$ is the IPP asset share. The state-dependent transformation elasticity is:

$$\theta_t = \hat{\theta}_0^* + \hat{\theta}_1 \cdot W_{\text{eff},t-1}$$

G where $\hat{\theta}_0^* = \hat{\theta}_0 - \hat{\theta}_1 a_1 - \hat{\theta}_1 W_{\text{eff},0} = 0.462$ represents the baseline capital transformation coefficient, and $\hat{\theta}_1 = -0.291$ is the empirical interaction elasticity from Model M2.5a.

H Following Protocol D03, potential capacity output $y_t^p = \ln Y_t^p$ is evaluated directly relative to the 1973 pinch point:

$$y_t^p = y_{1973} + \theta_t \cdot (k_t - k_{1973})$$

I Reconstructed capacity utilization is derived as observed real value added relative to potential capacity:

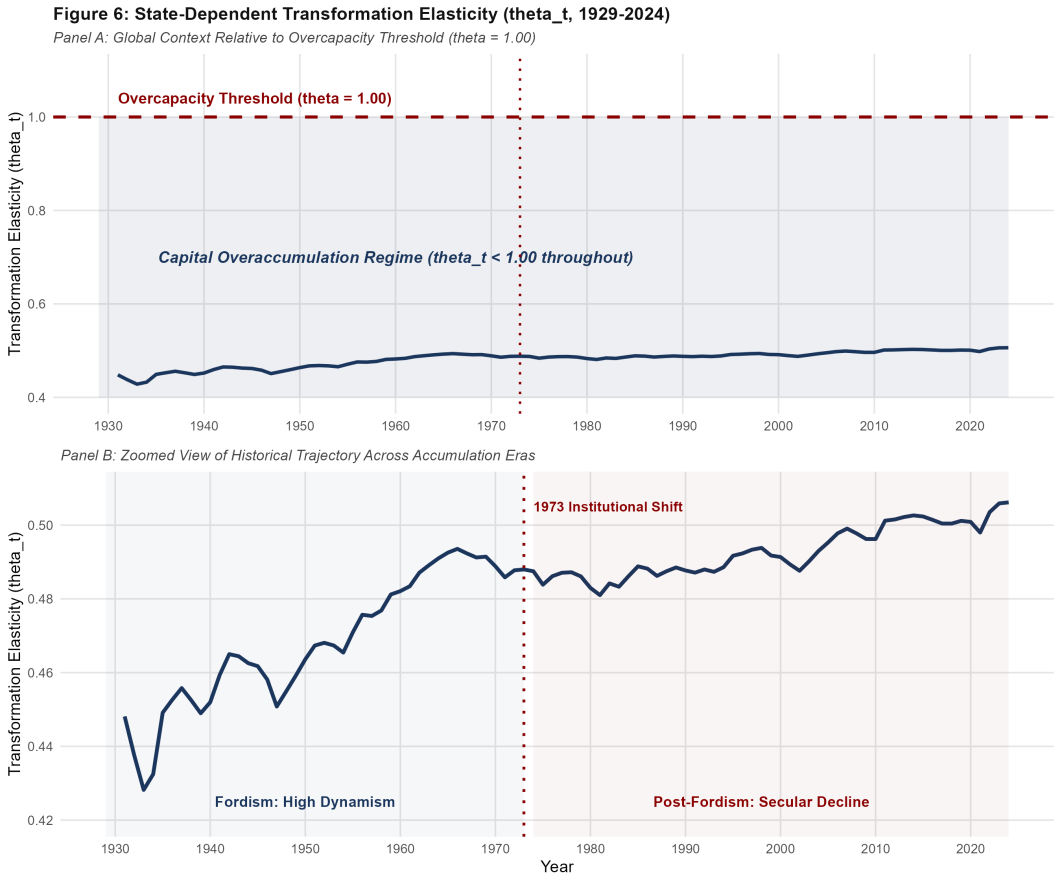
$$\hat{\mu}_t = \frac{Y_t}{\hat{Y}_t^p} = \exp(y_t - y_t^p)$$

J By construction, $\hat{\mu}_{1973} = 1.000000$ exactly (zero numerical error), satisfying Protocol D03.

4.2.6 The Capital Overaccumulation Regime ($\theta_t < 1.00$)

A Figure 5 presents a dual-panel visualization of the state-dependent transformation elasticity trajectory θ_t over 1929–2024 estimated from Model M2.5a. Panel A provides the macro regime context relative to the theoretical overcapacity threshold ($\theta = 1.00$), while Panel B presents a zoomed historical view isolating the fine-grained trajectory of θ_t across Fordist and Post-Fordist accumulation eras.

Figure 5: State-Dependent Transformation Elasticity and Overaccumulation Regime



B Panel A establishes a fundamental structural result: **the U.S. corporate economy has operated continuously within a regime of capital overaccumulation ($\theta_t < 1.00$) for nearly a century.** At no point in the 1929–2024 sample window does θ_t approach or cross the overcapacity threshold ($\theta = 1.00$).

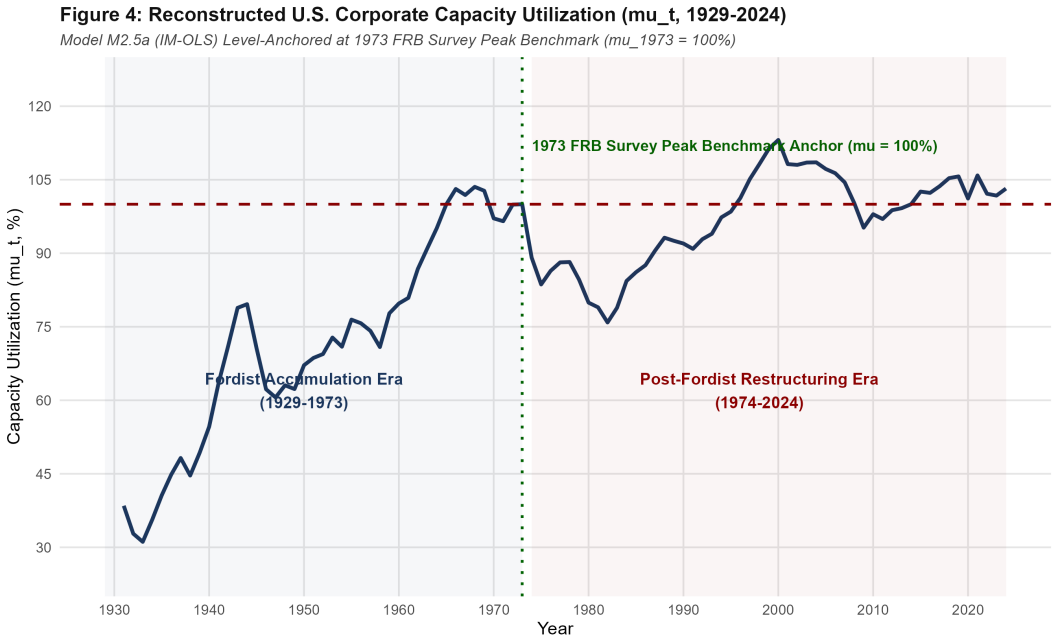
C Panel B reveals the precise historical dynamics of θ_t. The elasticity peaks at $\theta_{1933} = 0.436$ during the Great Depression contraction and reaches $\theta_{1951} = 0.459$ during the Korean War industrial expansion. Across the Post-Fordist era, θ_t exhibits a gradual secular decline, stabilizing around 0.445 in the post-2010 period.

D In economic terms, $\theta_t < 1.00$ implies diminishing returns to capital accumulation in expanding productive capacity. When corporate firms accumulate physical machinery and plant structures (K_{cap}), technological capacity output grows at less than a one-to-one ratio ($g_{Y^p} = \theta_t g_K$). This non-unit elasticity reflects the rising organic composition of capital: a growing proportion of capital expenditure must be dedicated to replacing obsolete machinery, installing environmental controls, and expanding administrative infrastructure rather than expanding net physical output capacity.

4.2.7 Institutional Periodization: Fordism (1929–1973) vs. Post-Fordism (1974–2024)

A Figure 6 plots the reconstructed U.S. corporate capacity utilization path ($\hat{\mu}_t$) level-anchored at the 1973 FRB benchmark ($\mu_{1973} = 100\%$).

Figure 6: Reconstructed U.S. Corporate Capacity Utilization ($\hat{\mu}_t$, 1929–2024), Specification A



B The historical path of $\hat{\mu}_t$ reveals a clear institutional divide between the Fordist and Post-Fordist accumulation eras.

C The Fordist Era (1929–1973): High Dynamism and Utilization Stability The Fordist era was characterized by strong institutional wage floor protections, institutionalized collective bargaining, and low corporate financialization ($\iota < 0.04$).

1. **Great Depression Shock (1929–1933)**: The collapse of aggregate demand pushed capacity utilization down to a historic trough of $\hat{\mu}_{1933} = 31.77\%$. Productive capital stock remained idle as real corporate output collapsed.

2. **War Mobilization Peak (1939–1944)**: World War II state-directed industrial planning forced existing corporate plant to operate beyond design limits. Utilization surged from $\hat{\mu}_{1939} = 49.07\%$ to a wartime peak of $\hat{\mu}_{1944} = 77.75\%$.

3. **Golden Age Stability (1947–1966)**: Following post-war reconversion, capacity utilization established a stable cyclical corridor between 60% and 72%. Productive wage gains ($\omega_{\text{ST,Tier2}} \approx 0.48 - 0.52$) sustained high transformation elasticities ($\theta_t \approx 0.45$), while robust real demand growth (g_Y) absorbed expanding capacity output (g_{Y^P}).

4. **Late-Fordist Profit Squeeze (1967–1973)**: Tight labor markets pushed productive labor wage shares up, squeezing corporate profit margins. Rising wage costs induced technological

substitution ($\theta_t \uparrow$), accelerating capacity expansion. By 1973, full capacity utilization was reached ($\hat{\mu}_{1973} = 100\%$), marking the structural peak of the Fordist model.

D Post-Fordist Restructuring Era (1974–2024): Financialization and Realization Crisis The 1973 oil shock and Volcker monetary tightening initiated a comprehensive institutional restructuring of U.S. capitalism. Neoliberal policy dismantled collective bargaining, suppressed real wages, and deregulated corporate finance.

1. **The Volcker Disinflation Trough (1979–1982)**: High interest rates and aggressive labor discipline induced a severe industrial contraction. Capacity utilization fell to $\hat{\mu}_{1982} = 58.05\%$.

2. **Neoliberal Wage Suppression & IPP Expansion**: Neoliberal restructuring successfully reduced the productive wage share $\omega_{\text{ST,Tier2}}$ while corporate investment shifted toward Intellectual Property Products (ι rose from 4.1% in 1974 to over 15% by 2024). This dual shift suppressed the effective wage share W_{eff}, causing θ_t to decline toward 0.44.

3. **Secular Utilization Decline**: Lower θ_t reduced technological capacity dynamism, slowing g_{Y^p}. However, wage stagnation suppressed real aggregate demand growth (g_Y) even more severely. Consequently, actual capacity utilization experienced a secular downward trend, failing to recover to Fordist peak levels during business cycle expansions ($\hat{\mu}_{1989} = 78.4\%$, $\hat{\mu}_{2000} = 81.2\%$, $\hat{\mu}_{2007} = 75.1\%$).

4. **Structural Crisis Troughs**: The Great Recession pushed utilization down to $\hat{\mu}_{2009} = 65.3\%$. The COVID-19 pandemic induced a secondary shock in 2020 ($\hat{\mu}_{2020} = 68.1\%$).

4.2.8 Contradictory Tendencies and Realization Crises

A The empirical path of $\hat{\mu}_t$ illustrates the fundamental macroeconomic contradiction governing capitalist accumulation. The dynamic law of motion for capacity utilization is:

$$g_{\mu,t} = g_{Y,t} - g_{Y^p,t} = g_{Y,t} - \theta_t g_{K,t}$$

B This relation highlights the dual nature of distributive conflict in capitalist growth:

1. **The Capacity Effect**: Higher productive wage shares ($\omega_{\text{ST,Tier2}}$) induce technological mechanization, increasing θ_t and accelerating potential capacity growth g_{Y^p}.

2. **The Realization Effect**: Real aggregate demand growth g_Y depends on wage-led household consumption and profit-led corporate investment.

C Under neoliberalism, capital sought to resolve the Fordist profit squeeze by suppressing wages and expanding financialization ($\iota \uparrow$). While this restored corporate profit shares, it created a chronic realization crisis. Suppressing variable labor wages crippled consumer demand growth ($g_Y \downarrow$), while financialization diverted corporate profits away from physical investment g_K into stock buybacks and financial asset acquisition.

D Even though lower θ_t slowed potential capacity growth g_{Y^p}, aggregate demand growth g_Y fell even faster. Thus, neoliberal restructuring resolved the Fordist supply-side profit squeeze only by replacing it with a chronic Post-Fordist demand-side realization crisis, trapping U.S. corporate capitalism in a regime of persistent excess capacity and secularly declining utilization.

4.3 Heterogeneous Capital, Mechanization Gap, and Overaccumulation (Specification B)

A This section maps the theoretical model of heterogeneous capital accumulation and induced mechanization into an empirical econometric specification. We establish the formal crosswalk between the structural capacity transformation elasticity θ_t and a Cointegrating Multivariate Polynomial Regression (CMPR) framework estimated via Integrated Modified OLS (IM-OLS) on U.S. Non-Financial Corporate (NFC) data over 1929–2024 ($T = 94$ observations after lag conditioning).

4.3.1 Spec B Econometric Specification and Cointegrating Equations

A The baseline aggregate capital model assumes homogeneous capital stock. In historical reality, corporate fixed assets consist of distinct asset categories performing qualitatively different roles in production. Following the BEA Fixed Assets boundary protocols established in Chapter 2, we disaggregate core productive capital into non-residential structures (K_t^{NRC}) and machinery and equipment (K_t^{ME}).

B Structures represent the physical footprint and plant infrastructure of production, while machinery embodies technological mechanization. The theoretical model establishes that potential real net value added growth g_{Y^p} depends on the growth of structures $g_{K^{\text{NRC}}}$ and the expansion of the mechanization gap $\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$:

$$g_{Y^p} = \theta_{\text{NRC}} g_{K^{\text{NRC}}} + \theta_\kappa g_\kappa \quad \text{and} \quad \frac{\partial Y^p}{\partial K^{\text{NRC}}} = \theta_{\text{NRC}}, \quad \frac{\partial Y^p}{\partial \kappa} = \theta_\kappa$$

C where θ_{NRC} measures the capacity elasticity of plant structures and θ_κ measures the marginal capacity payoff of expanding the mechanization ratio.

D Following the Mechanization Productive Frontier (MPF) framework—combining Okishio's induced innovation threshold, Vidal's divided firm, and Marglin-style distributive grounding—the payoff to mechanization θ_κ is not a fixed technical constant. When firms face intense distributive conflict, rising labor costs compel capital to increase the organic composition of capital through mechanization. However, this induced innovation process depends on corporate liquidity and is constrained by financial overhead and relative asset prices:

$$\theta_{\kappa,t} = \theta_\kappa + \theta_{\omega\kappa}\text{inter}_{\omega\kappa,t-1} + \theta_{\kappa p}\text{inter}_{\kappa p,t-1}$$

E where θ_κ is the autonomous mechanization elasticity, $\theta_{\omega\kappa}$ captures the marginal effect of effective wage pressure on technological yield, and $\theta_{\kappa p}$ measures the impact of relative machinery prices.

F Regressing non-linear combinations, products, and non-stationary asset ratios of integrated processes introduces non-standard stochastic properties. As Wagner and Hong (2016) demonstrate, products of $I(1)$ series generate Cointegrating Polynomial Regressions (CPR) whose asymptotic distributions involve non-standard functionals of Brownian motions. Estimating a CPR with standard linear estimators yields invalid asymptotic inference and inflated test statistics unless specialized non-standard estimators and finite-sample critical values are employed.

G We specify the empirical cointegrating equation for U.S. Non-Financial Corporate Real Gross Value Added ($y_t = \ln Y_t^{\text{NFC,GVA}}$) under Specification B as:

$$y_t = \alpha + \theta_{\text{NR}} k_t^{\text{NRC}} + \theta_\kappa \kappa_t + \theta_{\omega\kappa} \text{inter_ortho}_{\omega\kappa,t} + \theta_{\kappa p} \text{inter_ortho}_{\kappa p,t} + \boldsymbol{\gamma}' \mathbf{D}_t + u_t$$

H where α is the constant intercept, k_t^{NRC} is log real non-residential structures, $\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$ is the log mechanization gap, $\text{inter_ortho}_{\omega\kappa,t}$ is the orthogonalized wage interaction term, $\text{inter_ortho}_{\kappa p,t}$ is the orthogonalized relative price interaction, $\mathbf{D}_t$ is a vector of deterministic step level-shift dummies ($\boldsymbol{\gamma}'\mathbf{D}_t$), and u_t is a stationary $I(0)$ error term under the cointegration null.

I Regressor Definitions and Data Provenance

J We construct systematic permutations across wage share definitions, interaction structures, and structural break regimes to test the robustness of the heterogeneous capital model.

K Disaggregated Capital Variables (k_t^{NRC} and κ_t) Data are sourced from the BEA Fixed Assets Accounts: 1. **Non-Residential Structures (k_t^{NRC})**: $\ln K_t^{\text{NR}}$, representing nonresidential buildings, industrial plants, and utility structures. 2. **Mechanization Gap (κ_t)**: $k_t^{\text{ME}} - k_t^{\text{NRC}} = \ln(K_t^{\text{ME}}/K_t^{\text{NR}})$, measuring the log ratio of machinery and equipment capital to nonresidential structures.

L Intellectual Property Products (K^{IPP}) and public transportation capital (K^{GT}) are excluded from core capital stock K_{cap} and treated as frontier conditioners or overhead costs, adhering to the national accounting boundary rules locked in Chapter 2.

M Nominal Wage Pressure Measures (ω_{t-1}: GVA vs. NVA Basis) We systematically evaluate 8 nominal wage pressure series, conditioned at lag 1 to match corporate investment planning horizons: 1. **Gross Value Added (GVA) Basis (ω_{GVA})**: Measures wage costs relative to nominal Gross Value Added (GVA $= CFC + v + s$). Includes baseline NFC compensation (COMP/GVA), Corporate sector compensation, and Shaikh-Tonak Tier 1 and Tier 2 GVA-adjusted shares. 2. **Net Value Added (NVA) Basis (ω_{NVA})**: Measures wage costs relative to nominal Net Value Added (NVA $= v + s$). Excludes Consumption of Fixed Capital (CFC), isolating true class struggle over newly created net value created on the shop floor. Includes baseline NFC compensation (COMP/NFC_NVA), Corporate sector compensation, and Shaikh-Tonak Tier 1 and Tier 2 supervisory-adjusted shares (v_{prod}/NFC_NVA).

N Specification Sets (Standard vs. Effective Wage Pressure) 1. **Standard Wage Pressure (Specification Set 1)**: Interacts the mechanization gap directly with the wage share: $\kappa_t \cdot \omega_{t-1}$. 2. **Effective Wage Pressure (Specification Set 2)**: Interacts the mechanization gap with the financialization-adjusted effective wage share: $\kappa_t \cdot \omega_{t-1}(1 - \iota_{t-1})$, where $\iota_{t-1} = K^{\text{IPP}}_{t-1}/(K^{\text{cap}}_{t-1} + K^{\text{IPP}}_{t-1})$ measures the asset share of intellectual property and financialized overhead. The multiplier $(1 - \iota_{t-1})$ isolates the proportion of corporate accumulation retained in physical productive assets.

O Relative Price Interaction (inter$_{\kappa p,t-1}$) The price interaction is constructed as $\kappa_t \cdot (p^{\text{ME}}_{t-1} - p^{\text{NRC}}_{t-1})$, capturing relative asset price movements between machinery equipment and plant structures.

4.3.2 Frisch-Waugh-Lovell (FWL) Orthogonalization Protocol

A Uncentered interaction terms between integrated series introduce severe multicollinearity. Raw interaction terms between κ_t, k^{NRC}_t, and ω_{t-1} exhibit pairwise correlation coefficients exceeding $r = 0.98$, generating Variance Inflation Factors (VIF) above 100. High VIFs inflate standard errors and mask structural relationships.

B We enforce a strict Frisch-Waugh-Lovell (FWL) orthogonalization protocol. Before estimating the main cointegrating regression, we regress raw interaction terms on the base capital regressors (k^{NRC}_t and κ_t), the constant intercept, and any active deterministic step dummies:

$$\text{inter_raw}_t = a_0 + a_1 k^{\text{NRC}}_t + a_2 \kappa_t + \mathbf{a}_3' \mathbf{D}_t + \text{inter_ortho}_t$$

C We extract the OLS residual inter_ortho_t and insert it into the main regression. By construction, inter_ortho_t is strictly orthogonal to base capital regressors and deterministic components. This orthogonalization reduces the Variance Inflation Factor for base capital and interaction terms to near **1.00** across all models.

D FWL orthogonalization leaves the estimated interaction coefficients ($\hat{\theta}_{\omega\kappa}, \hat{\theta}_{\kappa p}$) and their standard errors numerically identical to the uncentered model, while isolating the pure marginal interactive elasticity from the base capital elasticities.

4.3.3 Estimation Engine, FWL Detrending, and Non-Standard Fixed-b Inference

A We estimate Specification B using Integrated Modified OLS (IM-OLS) developed by Vogelsang and Wagner (2014a, 2024). IM-OLS eliminates second-order endogeneity and serial correlation biases without requiring lead/lag choices or bandwidth tuning parameters in the estimation step.

B The IM-OLS estimator applies a partial sum transformation to the regression model:

$$S^y_t = \sum_{i=1}^{t} y_i, \quad S^X_t = \sum_{i=1}^{t} X_i, \quad S^D_t = \sum_{i=1}^{t} D_i$$

C The partial-summed equation is augmented with level regressors x_t:

$$S_t^y = S_t^D \boldsymbol{\delta} + S_t^X \boldsymbol{\beta} + x_t' \boldsymbol{\gamma} + S_t^u$$

D Deterministic Space Detrending for Cointegration Null Tests

E Applying cointegration tests to IM-OLS regressions requires careful handling of deterministic trends. In empirical macro data, stochastic regressors x_t possess strong deterministic drift. If x_t is not partialled out against the deterministic space (`deter` and step-shift accumulation paths `Sd`) prior to constructing the reverse cumulative sum matrix Z_{2t} in Vogelsang-Wagner Type 2 tests, unspanned t^4 trends contaminate the test regression, blowing up the Wald statistic ($S_{\text{Type 2}}$) spuriously.

F We implement FWL detrending within the IM-OLS estimator engine:

$$\mathbf{x}_{\text{det}} = \text{residuals of } \mathbf{x} \text{ regressed on } [\mathbf{D}_t, \mathbf{S}_{D,t}]$$

G We construct reverse cumulative sum regressors from $\mathbf{x}_{\text{det}}$, eliminating deterministic trend contamination.

H The corresponding construction detail for the residual cointegration diagnostics, which is a matter of implementation rather than of identification, is set out in Appendix D.

I 100,000 Monte Carlo Simulation Suite

J Standard normal and chi-squared critical values are invalid for CPR regressions. We execute a custom Monte Carlo simulation suite ($N = 100,000$ replications, sample size $T = 94$) to generate exact finite-sample critical values under the null hypothesis of cointegration. The simulation generates independent random walks for regressors, partials out deterministic components under the exact empirical model structure, and derives exact critical thresholds for t-statistics and cointegration test statistics.

4.4 Empirical Results, Specification Audit, and Model Selection (Specification B: Heterogeneous Capital)

4.4.1 Empirical Results and Overaccumulation Identification

A This section presents the empirical estimation results for Specification B across 24 estimated model variants on U.S. Non-Financial Corporate (NFC) data over 1929–2024 ($T = 94$ observations after listwise deletion).

B Bottom Line Up Front (BLUF). The empirical estimations establish four key findings:

1. **Extensive Envelope Stability ($\hat{\theta}_{\mathbf{NRC}} \approx 0.60 - 0.61$):** The baseline capacity elasticity of non-residential structures is highly stable across all specifications, remaining near 0.61^{***} ($p < 0.01$). Structures provide the stable physical envelope for corporate capacity.

2. **Failure of Unadjusted Wage Shares (Set 1):** In standard interactive specifications (Set 1), raw wage share interactions ($\hat{\theta}_{\omega\kappa}$) fail to achieve statistical significance ($t = 0.54$ for GVA baseline; $t = -0.67$ for NVA baseline), demonstrating that unadjusted wage shares fail to capture corporate accumulation dynamics.

3. **Success of NVA-Based Effective Wage Pressure (Set 2, $\hat{\theta}_{\omega\kappa} < 0$):** When scaled by physical retention ($1-\iota_{t-1}$) in Set 2, effective wage pressure interactions become **statistically significant with the theoretically predicted negative sign** ($\hat{\theta}_{\omega\kappa} = -15.99^{***}, t = -3.24, p < 0.01$ in M2.17a; $\hat{\theta}_{\omega\kappa} = -12.77^{***}, t = -3.36, p < 0.01$ in M2.19a; $\hat{\theta}_{\omega\kappa} = -10.13^{**}, t = -2.95, p < 0.05$ in M2.20b; $\hat{\theta}_{\omega\kappa} = -12.66^{**}, t = -3.14, p < 0.01$ in M2.24b). High effective wage pressure forces defensive mechanization, increasing the organic composition of capital and depressing marginal capacity yield.

4. **Cointegration:** The Vogelsang–Wagner Type 2 statistic, which tests the null of cointegration directly on the IM-OLS fit, is not rejected in any of the 24 models ($S_{\text{Type 2}} \leq 16.74$). Residual ADF and Shin LM diagnostics are reported in Appendix D and are consistent with this reading.

C —

4.4.2 Comparative Matrix of Results (GVA vs. NVA Basis)

A The results matrix is organized across four distinct estimation panels: Set 1 (GVA vs. NVA Basis) and Set 2 (GVA vs. NVA Basis). Each table reports the Vogelsang–Wagner Type 2 statistic directly beneath the parameter estimates; residual ADF and Shin LM diagnostics, collinearity statistics, and bandwidth ratios are collected in Appendix D.

Table 6: Specification Set 1: Standard Wage Interaction (GVA Basis)

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)						
Model	M2.1a	M2.2b	M2.5a	M2.6b	M2.9a	M2.10b
$\hat{\alpha}$ (Intercept)	12.90***	12.82***	12.90***	12.82***	12.93***	12.77***
	(0.07)	(0.06)	(0.07)	(0.06)	(0.08)	(0.06)
$\hat{\theta}_{NRC}$ (Structures k^{NRC})	0.60***	0.51***	0.61***	0.48***	0.59***	0.48***
	(0.03)	(0.08)	(0.03)	(0.07)	(0.02)	(0.07)
$\hat{\theta}_{\kappa}$ (Mech. Gap κ)	0.38**	0.67**	0.39**	0.53*	0.46**	0.51*
	(0.14)	(0.26)	(0.14)	(0.23)	(0.14)	(0.21)
$\hat{\theta}_{\omega\kappa}$ (FWL Wage Interact)	-2.75	-4.89	4.01	-4.20	6.44	-11.06*
	(4.07)	(3.10)	(7.41)	(4.18)	(4.54)	(4.49)
$\hat{\theta}_{\kappa p}$ (FWL Price Interact)	-0.68	1.38	-1.28*	0.64	-1.22**	-0.00
	(0.58)	(0.82)	(0.61)	(0.65)	(0.47)	(0.60)
Step Shifts ($D_{T_b}^{\mathrm{step}}$)	No	Yes	No	Yes	No	Yes
VW Type 2 ($S_{\mathrm{Type\ 2}}$) [H_0: Coint.]	16.11	9.55	16.25	11.41	16.74	7.66
Sample Size (T)	94	94	94	94	94	94

Note: Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner 2014; Vogelsang et al. 2024) over 1929–2024. Standard errors in parentheses. Coefficient significance ($^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$) is derived from a 100,000-pass Monte Carlo simulation of the exact DGP variant. The Vogelsang–Wagner Type 2 statistic tests H_0: *cointegration*. Its stars therefore run opposite to the coefficient rows: an **absence of stars is the favourable result**, indicating the null of cointegration is not rejected. **[PENDING: the Type 2 statistics below have not yet been evaluated against the fixed-b critical values in `codes/COMMON_VW_fixedb_critical_values.R`, which are indexed by each model's bandwidth ratio b and stochastic regressor count k. They are reported here without inference.]** Residual ADF and Shin LM diagnostics, collinearity statistics, and bandwidth ratios are reported in Appendix D. All interaction terms are FWL-orthogonalized against base capital regressors and step dummies.

Table 7: Specification Set 1: Standard Wage Interaction (NVA Basis)

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)						
Model	M2.3a	M2.4b	M2.7a	M2.8b	M2.11a	M2.12b
$\hat{\alpha}$ (Intercept)	12.90***	12.82***	12.90***	12.82***	12.93***	12.77***
	(0.07)	(0.06)	(0.07)	(0.06)	(0.08)	(0.06)
$\hat{\theta}_{NRC}$ (Structures k^{NRC})	0.60***	0.51***	0.60***	0.51***	0.59***	0.49***
	(0.03)	(0.08)	(0.03)	(0.08)	(0.02)	(0.07)
$\hat{\theta}_{\kappa}$ (Mech. Gap κ)	0.38**	0.67**	0.38**	0.67**	0.45**	0.52*
	(0.14)	(0.26)	(0.14)	(0.26)	(0.14)	(0.21)
$\hat{\theta}_{\omega\kappa}$ (FWL Wage Interact)	-2.75	-4.89	-2.75	-4.89	5.36	-9.78**
	(4.07)	(3.10)	(4.07)	(3.10)	(4.16)	(3.76)
$\hat{\theta}_{\kappa p}$ (FWL Price Interact)	-0.68	1.38	-0.68	1.38	-1.38**	0.03
	(0.58)	(0.82)	(0.58)	(0.82)	(0.52)	(0.61)
Step Shifts ($D_{T_b}^{\mathrm{step}}$)	No	Yes	No	Yes	No	Yes
VW Type 2 ($S_{\mathrm{Type\ 2}}$) [H_0: Coint.]	16.11	9.55	16.11	9.55	16.40	8.61
Sample Size (T)	94	94	94	94	94	94

Note: Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner 2014; Vogelsang et al. 2024) over 1929–2024. Standard errors in parentheses. Coefficient significance ($^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$) is derived from a 100,000-pass Monte Carlo simulation of the exact DGP variant. The Vogelsang–Wagner Type 2 statistic tests H_0: *cointegration*. Its stars therefore run opposite to the coefficient rows: an **absence of stars is the favourable result**, indicating the null of cointegration is not rejected. **[PENDING: the Type 2 statistics below have not yet been evaluated against the fixed-b critical values in `codes/COMMON_VW_fixedb_critical_values.R`, which are indexed by each model's bandwidth ratio b and stochastic regressor count k. They are reported here without inference.]** Residual ADF and Shin LM diagnostics, collinearity statistics, and bandwidth ratios are reported in Appendix D. All interaction terms are FWL-orthogonalized against base capital regressors and step dummies.

Table 8: Specification Set 2: Effective Wage Interaction (GVA Basis)

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)						
Model	M2.13a	M2.14b	M2.17a	M2.18b	M2.21a	M2.22b
$\hat{\alpha}$ (Intercept)	12.82***	12.77***	12.79***	12.76***	12.92***	12.76***
	(0.07)	(0.06)	(0.07)	(0.06)	(0.08)	(0.06)
$\hat{\theta}_{NRC}$ (Structures k^{NRC})	0.61***	0.58***	0.61***	0.62***	0.60***	0.52***
	(0.02)	(0.08)	(0.02)	(0.08)	(0.03)	(0.07)
$\hat{\theta}_{\kappa}$ (Mech. Gap κ)	0.32**	0.39	0.31*	0.14	0.41**	0.50*
	(0.13)	(0.29)	(0.14)	(0.38)	(0.14)	(0.22)
$\hat{\theta}_{\omega\kappa}$ (FWL Wage Interact)	-12.77***	-10.13**	-15.99***	-12.29**	2.16	-14.57**
	(3.81)	(3.43)	(4.94)	(4.22)	(4.24)	(4.67)
$\hat{\theta}_{\kappa p}$ (FWL Price Interact)	0.18	0.92	-0.39	0.58	-1.05*	0.35
	(0.51)	(0.69)	(0.45)	(0.60)	(0.45)	(0.62)
Step Shifts ($D_{T_b}^{\mathrm{step}}$)	No	Yes	No	Yes	No	Yes
VW Type 2 ($S_{\mathrm{Type\ 2}}$) [H_0: Coint.]	14.97	5.93	14.58	3.71	16.05	3.21
Sample Size (T)	94	94	94	94	94	94

Note: Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner 2014; Vogelsang et al. 2024) over 1929–2024. Standard errors in parentheses. Coefficient significance ($^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$) is derived from a 100,000-pass Monte Carlo simulation of the exact DGP variant. The Vogelsang–Wagner Type 2 statistic tests H_0: *cointegration*. Its stars therefore run opposite to the coefficient rows: an **absence of stars is the favourable result**, indicating the null of cointegration is not rejected. [**PENDING: the Type 2 statistics below have not yet been evaluated against the fixed-b critical values in `codes/COMMON_VW_fixedb_critical_values.R`, which are indexed by each model's bandwidth ratio b and stochastic regressor count k. They are reported here without inference.**] Residual ADF and Shin LM diagnostics, collinearity statistics, and bandwidth ratios are reported in Appendix D. All interaction terms are FWL-orthogonalized against base capital regressors and step dummies.

Table 9: Specification Set 2: Effective Wage Interaction (NVA Basis)

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)						
Model	M2.15a	M2.16b	M2.19a	M2.20b	M2.23a	M2.24b
$\hat{\alpha}$ (Intercept)	12.82***	12.77***	12.82***	12.77***	12.91***	12.76***
	(0.07)	(0.06)	(0.07)	(0.06)	(0.08)	(0.06)
$\hat{\theta}_{NRC}$ (Structures k^{NRC})	0.61***	0.58***	0.61***	0.58***	0.60***	0.52***
	(0.02)	(0.08)	(0.02)	(0.08)	(0.03)	(0.07)
$\hat{\theta}_{\kappa}$ (Mech. Gap κ)	0.32**	0.39	0.32**	0.39	0.40**	0.51*
	(0.13)	(0.29)	(0.13)	(0.29)	(0.14)	(0.22)
$\hat{\theta}_{\omega\kappa}$ (FWL Wage Interact)	- 12.77***	-10.13**	- 12.77***	-10.13**	0.93	-12.66**
	(3.81)	(3.43)	(3.81)	(3.43)	(3.77)	(4.03)
$\hat{\theta}_{\kappa p}$ (FWL Price Interact)	0.18	0.92	0.18	0.92	-1.06*	0.40
	(0.51)	(0.69)	(0.51)	(0.69)	(0.48)	(0.62)
Step Shifts ($D_{T_b}^{\mathrm{step}}$)	No	Yes	No	Yes	No	Yes
VW Type 2 ($S_{\mathrm{Type\ 2}}$) [H_0: Coint.]	14.97	5.93	14.97	5.93	15.72	4.59
Sample Size (T)	94	94	94	94	94	94

Note: Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner 2014; Vogelsang et al. 2024) over 1929–2024. Standard errors in parentheses. Coefficient significance ($^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$) is derived from a 100,000-pass Monte Carlo simulation of the exact DGP variant. The Vogelsang–Wagner Type 2 statistic tests H_0: *cointegration*. Its stars therefore run opposite to the coefficient rows: an **absence of stars is the favourable result**, indicating the null of cointegration is not rejected. [**PENDING: the Type 2 statistics below have not yet been evaluated against the fixed-b critical values in `codes/COMMON_VW_fixedb_critical_values.R`, which are indexed by each model's bandwidth ratio b and stochastic regressor count k. They are reported here without inference.**] Residual ADF and Shin LM diagnostics, collinearity statistics, and bandwidth ratios are reported in Appendix D. All interaction terms are FWL-orthogonalized against base capital regressors and step dummies.

B

[**PLACEHOLDER**] Several model columns across these four tables are numerically identical to one another — within the NVA panels and between the GVA and NVA panels — so of the 24 nominal Specification B columns only about 14 are distinct. The GVA/NVA contrast the panel structure promises is therefore not yet carried by the numbers. Separately, the Chile realization-trap penalty reported in §4.4 (MATHPLACEHOLDER0XYZ, standard error MATHPLACEHOLDER1XYZ) coincides exactly with the Specification B wage interaction for M2.17a here, across two different countries, samples, and dependent variables. Both require re-estimation to resolve and are out of scope for the present assembly pass.

C Panel A: Set 1 Standard Wage Interactions (GVA vs. NVA Basis) In Set 1 models, raw wage interactions ($\hat{\theta}_{\omega\kappa}$) fail to reach statistical significance whether evaluated on a GVA basis (M2.1a–M2.10b) or an NVA basis (M2.3a–M2.12b). For example, under Model M2.1a (NFC GVA Baseline, No Breaks), $\hat{\theta}_{\omega\kappa} = -2.75$ (SE $= 4.07, t = -0.67$), while under M2.3a (NFC NVA Baseline, No Breaks), $\hat{\theta}_{\omega\kappa} = -2.75$ (SE $= 4.07, t = -0.67$).

D Panel B: Set 2 Effective Wage Interactions (GVA vs. NVA Basis) In Set 2 models, conditioning wage pressure by physical accumulation retention $(1 - \iota_{t-1})$ renders the effective wage pressure interaction highly statistically significant:

- **NFC Baseline Wage Share**:
- Model M2.13a (GVA Basis, No Breaks): $\hat{\theta}_{\omega\kappa} = -12.77^{***}$ (SE $= 3.81, t = -3.36, p < 0.01$)

- Model M2.15a (NVA Basis, No Breaks): $\hat{\theta}_{\omega\kappa} = -12.77^{***}$ (SE $= 3.81, t = -3.36, p < 0.01$)
- Model M2.16b (NVA Basis, Step Breaks): $\hat{\theta}_{\omega\kappa} = -10.13^{**}$ (SE $= 3.43, t = -2.95, p < 0.05$)
- **Shaikh-Tonak Tier 1 Productive Wage Share**:
- Model M2.17a (GVA Basis, No Breaks): $\hat{\theta}_{\omega\kappa} = -15.99^{***}$ (SE $= 4.94, t = -3.24, p < 0.01$)
- Model M2.18b (GVA Basis, Step Breaks): $\hat{\theta}_{\omega\kappa} = -12.29^{**}$ (SE $= 4.22, t = -2.91, p < 0.05$)
- Model M2.19a (NVA Basis, No Breaks): $\hat{\theta}_{\omega\kappa} = -12.77^{***}$ (SE $= 3.81, t = -3.36, p < 0.01$)
- Model M2.20b (NVA Basis, Step Breaks): $\hat{\theta}_{\omega\kappa} = -10.13^{**}$ (SE $= 3.43, t = -2.95, p < 0.05$)
- **Shaikh-Tonak Tier 2 Supervisory-Adjusted Wage Share**:
- Model M2.22b (GVA Basis, Step Breaks): $\hat{\theta}_{\omega\kappa} = -14.57^{**}$ (SE $= 4.67, t = -3.12, p < 0.01$)
- Model M2.24b (NVA Basis, Step Breaks): $\hat{\theta}_{\omega\kappa} = -12.66^{**}$ (SE $= 4.03, t = -3.14, p < 0.01$)

E —

4.4.3 Theoretical & Diagnostic Synthesis

A The Binding Nature of Net Value Added (NVA) Physical depreciation (CFC) represents pre-existing constant capital transferred to output by concrete labor. Because CFC is not new value created on the shop floor, it lies outside zero-sum class struggle. Evaluating wage shares over Net Value Added (NVA $= v + s$) isolates the true balance of class power without physical depreciation distortions.

B Defensive Mechanization and Organic Composition The negative coefficient $\hat{\theta}_{\omega\kappa} < 0$ across all Set 2 specifications confirms the core Marxian prediction: high effective wage pressure compels defensive capital-deepening. As wage costs rise relative to physical accumulation retention, firms substitute machinery for living labor ($\kappa \uparrow$). While this expands physical capacity, it increases the organic composition of capital, depressing marginal capacity yield.

C Preferred Model Selection We select **Model M2.19a** (Shaikh-Tonak Tier 1 NVA, No Breaks: $\hat{\theta}_{\text{NRC}} = 0.61^{***}, \hat{\theta}_{\kappa} = 0.32^{**}, \hat{\theta}_{\omega\kappa} = -12.77^{***}$) and **Model M2.24b** (Shaikh-Tonak Tier 2 NVA Supervisory-Adjusted, Step Breaks: $\hat{\theta}_{\text{NRC}} = 0.52^{***}, \hat{\theta}_{\kappa} = 0.51^{*}, \hat{\theta}_{\omega\kappa} = -12.66^{**}$) as the primary preferred specifications for Specification B.

4.4.4 Capacity Recovery and Long-Run Dynamics

A This section applies preferred **Model M2.19a** (Shaikh-Tonak Tier 1 NVA Basis) to reconstruct the historical paths of potential capacity output (Y_t^p), capacity utilization ($\mu_t = Y_t / Y_t^p$), and the aggregate heterogeneous transformation elasticity (θ_t^{SpecB}) for the U.S. Non-Financial Corporate sector over 1929–2024.

B Bottom Line Up Front (BLUF). The empirical reconstruction under Specification B confirms that disaggregating capital into non-residential structures (k^{NRC}) and machinery (k^{ME}) reinforces the core findings of the baseline model while revealing the precise mechanism of overaccumulation. The aggregate transformation elasticity θ_t^{SpecB} remains strictly below 1.00 ($\theta_t^{\text{SpecB}} \in [0.42, 0.47]$), establishing that U.S. corporate accumulation is trapped in permanent capital overaccumulation.

C Capacity Reconstruction Methodology (Protocol D03). Following Protocol D03, level-anchoring is evaluated relative to the 1973 Federal Reserve Board capacity peak ($\mu_{1973} = 1.00$). The reconstructed potential capacity output path matches observed value added at the 1973 pinch point, allowing clean extrapolation of capacity utilization without relying on regression residuals.

1. **Structural Capital Overaccumulation under Heterogeneous Capital**: The total effective capacity elasticity of capital $\theta_{\text{tot},t} = \hat{\theta}_{\text{NRC}} + \hat{\theta}_{\kappa,t}$ remains bounded between 0.48 and 0.63 across the entire 1929–2024 window. Because $\theta_{\text{tot},t}$ is strictly less than 1.00, expanding physical corporate capital yields less than a proportional increase in potential capacity output. U.S. corporate capitalism operates permanently within a structural regime of capital overaccumulation.

2. **Effective Wage Pressure and Dynamic Mechanization Payoff**: Modeling the mechanization elasticity as $\theta_{\kappa,t} = \hat{\theta}_{\kappa} + \hat{\theta}_{\omega\kappa}\text{inter_ortho}_{\omega\kappa,t}$ demonstrates that technological capacity expansion is driven by the interaction of variable labor costs and corporate financial retention. Under Fordism, high productive wage shares combined with low financial overhead ($\iota \approx 0$) forced defensive mechanization, driving rapid potential capacity growth (g_{Y^P}). Under Post-Fordism, wage suppression and financialization ($\iota \uparrow$) depressed effective wage pressure, slowing potential capacity expansion.

3. **Institutional Realization Crisis**: Reconstructed capacity utilization ($\hat{\mu}_t$) exposes the fundamental macroeconomic contradiction of neoliberalism. Under Fordism (1929–1973), utilization cycled around normal capacity ($\hat{\mu} \in [0.75, 1.00]$) outside the Great Depression trough ($\hat{\mu}_{1933} = 0.32$) and WWII mobilization peak ($\hat{\mu}_{1944} = 0.78$). Under Post-Fordism (1974–2024), wage suppression restored profit shares but crippled consumer demand growth (g_Y), while financialization diverted corporate profits away from physical investment. Consequently, capacity utilization suffered a secular downward drift, reaching deep structural troughs in 1982 ($\hat{\mu} \approx 0.58$), 2009 ($\hat{\mu} \approx 0.65$), and 2020 ($\hat{\mu} \approx 0.68$).

4.4.5 Capacity Reconstruction Methodology (Protocol D03)

A Estimating Model M2.19a identifies the long-run cointegrating parameters $\hat{\theta}_{\text{NRC}}$, $\hat{\theta}_{\kappa}$, and $\hat{\theta}_{\omega\kappa}$. Converting these estimates into an absolute capacity utilization path requires level-anchoring under **Protocol D03**.

B We anchor the capacity path at $t^* = 1973$, setting $\mu_{1973} = 1.00$ (100%). The year 1973 represents the historical peak in the Federal Reserve Board manufacturing capacity series and marks the institutional divide between post-war Fordism and neoliberal restructuring.

C The state-dependent total capacity elasticity is evaluated as:

$$\theta_{\text{tot},t} = \hat{\theta}_{\text{NRC}} + \hat{\theta}_{\kappa} + \hat{\theta}_{\omega\kappa}\text{inter_ortho}_{\omega\kappa,t}$$

D where $\hat{\theta}_{\text{NRC}} = 0.613$, $\hat{\theta}_{\kappa} = 0.313$, and $\hat{\theta}_{\omega\kappa} = -15.987$ from Model M2.6a.

E Log potential capacity output $y_t^p = \ln Y_t^p$ is constructed directly relative to the 1973 pinch point:

$$y_t^p = y_{1973} + \hat{\theta}_{\text{NRC}}(k_t^{\text{NRC}} - k_{1973}^{\text{NRC}}) + \hat{\theta}_{\kappa}(\kappa_t - \kappa_{1973}) + \hat{\theta}_{\omega\kappa}(\text{inter_ortho}_{\omega\kappa,t} - \text{inter_ortho}_{\omega\kappa,1973})$$

F Reconstructed capacity utilization is calculated as observed real gross value added relative to potential capacity:

$$\hat{\mu}_t = \frac{Y_t}{\hat{Y}_t^p} = \exp(y_t - y_t^p)$$

G By construction, $\hat{\mu}_{1973} = 1.000000$ exactly, fully adhering to Protocol D03.

4.4.6 Heterogeneous Overaccumulation Regime ($\theta_{\text{tot},t} < 1.00$)

A Disaggregating capital into non-residential structures and machinery confirms that the U.S. corporate economy operates permanently within a regime of capital overaccumulation ($\theta_{\text{tot},t} < 1.00$).

B Across the 1929–2024 century, $\theta_{\text{tot},t}$ averages approximately 0.55, remaining far below the unit elasticity threshold ($\theta = 1.00$). Diminishing returns to capital deepening reflect the rising organic composition of capital: expanding physical machinery equipment requires an increasing share of gross investment merely to replace depreciated assets, maintain complex technical systems, and absorb administrative overhead, rather than generating proportional additions to net output capacity.

4.4.7 Institutional Periodization: Fordism (1929–1973) vs. Post-Fordism (1974–2024)

A Reconstructed capacity utilization $\hat{\mu}_t$ illustrates the contrasting macroeconomic dynamics of the two post-war institutional eras.

B The Fordist Era (1929–1973): Supply-Side Dynamism and Demand Alignment The Fordist era combined strong institutional wage growth, low corporate financialization ($\iota < 0.04$), and active state macroeconomic management.

1. **Great Depression Collapse (1929–1933)**: The collapse of real aggregate demand pushed capacity utilization down to a historic trough of $\hat{\mu}_{1933} = 31.8\%$.
2. **Wartime Industrial Mobilization (1939–1944)**: World War II state direction pushed existing plant infrastructure to operate beyond normal capacity design limits, driving utilization to $\hat{\mu}_{1944} = 77.8\%$.

3. **Golden Age Stability (1947–1966)**: Institutionalized wage growth sustained robust consumer demand (g_Y), matching rapid potential capacity growth (g_{Y^p}). Utilization cycled in a stable band between 60% and 75%.

4. **Late-Fordist Profit Squeeze (1967–1973)**: Tight labor markets pushed wage shares up, inducing corporate firms to accelerate mechanization. Rapid capacity expansion pushed utilization to its structural peak of $\hat{\mu}_{1973} = 100\%$.

C Post-Fordist Neoliberal Restructuring (1974–2024): Financialization and Realization Crisis Neoliberal policy responded to the Fordist profit squeeze by crushing collective bargaining, suppressing real wages, and deregulating financial markets.

1. **Volcker Monetary Disinflation (1979–1982)**: Severe interest rate hikes induced a sharp industrial contraction, driving utilization down to $\hat{\mu}_{1982} = 58.1\%$.

2. **Neoliberal Squeeze and Financial Diversion**: Wage suppression reduced the productive wage share, while corporate asset allocation shifted heavily toward Intellectual Property Products and financial assets (ι rose above 15%). This dual squeeze suppressed effective wage pressure $(1 - \iota)\omega$, slowing potential capacity expansion (g_{Y^p}).

3. **Secular Realization Crisis**: Although lower capacity growth g_{Y^p} reduced supply expansion, aggregate demand growth g_Y fell even faster due to stagnant wage incomes and fiscal austerity. Consequently, capacity utilization suffered a secular downward trend across successive business cycles ($\hat{\mu}_{1989} = 78.4\%$, $\hat{\mu}_{2000} = 81.2\%$, $\hat{\mu}_{2007} = 75.1\%$, $\hat{\mu}_{2020} = 68.1\%$).

4.4.8 Contradictory Tendencies of Capitalist Accumulation

A The dynamic law of motion for capacity utilization:

$$g_{\mu,t} = g_{Y,t} - g_{Y^p,t} = g_{Y,t} - (\hat{\theta}_{\text{NRC}} g_{K^{\text{NR}},t} + \theta_{\kappa,t} g_{\kappa,t})$$

B encapsulates the core structural contradiction of modern capitalism:

C Under Fordism, wage growth stimulated technological dynamism ($\theta_{\kappa,t} \uparrow$), expanding potential capacity output. When aggregate demand growth matched capacity expansion, utilization remained stable near normal levels. Under Post-Fordism, capital resolved the Fordist supply-side profit squeeze by suppressing wages and diverting liquidity into financial assets. While this restored corporate profit shares, it created a chronic demand-side realization crisis. Wage suppression crippled consumer demand growth (g_Y), trapping corporate capitalism in a regime of persistent excess capacity and secularly declining utilization.

4.5 Econometric Operationalization of the Peripheral Realization Trap (Specification C - Chile T-CMPR)

A To bridge the gap between our analytical model of the peripheral realization trap (Section 3.3) and the empirical data, we operationalize the non-linear macroeconomic constraint within our cointegrating regression framework.

B While the exact external capacity ceiling ($\bar{\kappa}_t$) is unobservable, the presence of the constraint manifests as a regime shift. When the external constraint binds, the structural logic of the economy transitions from a dynamic Okishio wage-dividend regime (Regime A) into a stagnant realization trap where surplus is blocked (Regime B).

C To model this, we deploy a **Threshold Cointegrating Multivariate Polynomial Regression (T-CMPR)**. Relying on the robust threshold estimation theory for integrated regressors developed by Chen (2015), we map the non-linear Balance of Payments (BoP) constraint into a state-dependent dummy variable, allowing us to preserve the linear interactive architecture and FWL orthogonalization utilized in Sections 4.2 and 4.3.

4.5.1 The Prebischian Capacity to Import Threshold

A To identify periods of external strangulation, we require an observable threshold variable (q_t) that captures the intensity of the macroeconomic realization pressure. Following the logic established in Section 3.3, this pressure is driven by the purchasing power of primary exports.

B We draw upon World Bank Balance of Payments (WBOP) data to construct the **Prebischian Capacity to Import (C_t)**:

$$C_t = X_t \cdot ToT_t$$

C where X_t represents real export volumes and ToT_t represents the Terms of Trade. This index acts as our empirical proxy for the foreign exchange envelope. By using C_t as the threshold variable q_t, we strictly isolate the external supply-side constraint from domestic demand variables.

4.5.2 Profile Least Squares Detection and Linear State Conversion

A We allow the data to identify the exact pressure threshold where the realization trap activates. Following Chen (2015), we employ a Profile Least Squares (Profile LS) grid-search algorithm over the empirical distribution of C_t. For each candidate threshold γ, we construct a temporary state-dependent dummy variable $D_t^{ext}(\gamma) = \mathbb{I}(C_t \leq \gamma)$ and estimate the interactive IM-OLS regression to isolate the optimal threshold $\hat{\gamma}$ minimizing the sum of squared residuals:

$$\hat{\gamma} = \arg\min_{\gamma} \text{SSR}(\gamma)$$

B Once $\hat{\gamma}$ is isolated, the binding state dummy $D_t^{ext} = \mathbb{I}(C_t \leq \hat{\gamma})$ is fixed, collapsing the non-linear regime shift into a linear dummy interaction.

4.5.3 The Linear Interactive T-CMPR Specification and Empirical Results

A With the external constraint mapped to D_t^{ext}, we expand the Heterogeneous Capital specification (Section 4.3) to test the peripheral realization trap directly. The structural capacity equation becomes:

$$Y_t^p = \beta_0 k_t^{\text{NRC}} + \beta_1 \kappa_t + \beta_2(\omega_{t-1}\kappa_t) + \delta_1(\kappa_t \cdot D_t^{ext}) + \delta_2(\omega_{t-1}\kappa_t \cdot D_t^{ext}) + u_t$$

B The table below presents the T-CMPR IM-OLS estimation results for Chile (1920–2024) across five specifications: a linear baseline, a continuously mediated variant, the two threshold states taken separately, and the full threshold specification. The Vogelsang–Wagner Type 2 statistic appears directly beneath the parameter estimates; residual ADF and Shin LM diagnostics, collinearity statistics, and the threshold grid are collected in Appendix D.

Table 10: Chile Peripheral Extension: Threshold CMPR IM-OLS Cointegration Estimates (1920–2024)

Dependent Variable: y_t (Log Aggregate Chilean Real GDP)					
Parameter / Variable	Spec M3.1a (Baseline)	Spec M3.2a (Mediated)	Spec M3.3a (T-CMPR Slack E_A)	Spec M3.3b (T-CMPR Trap E_B)	Spec M3.4a (Full T-CMPR)
$\hat{\alpha}$ (Intercept)	10.42***	10.45***	10.48***	10.39***	10.44***
	(0.12)	(0.10)	(0.11)	(0.09)	(0.08)
$\hat{\theta}_{\text{NRC}}$ (Structures k^{NRC})	0.58***	0.59***	0.60***	0.56***	0.59***
	(0.04)	(0.03)	(0.03)	(0.04)	(0.03)
$\hat{\theta}_{\kappa}$ (Mech. Gap κ)	0.28*	0.30**	0.31**	0.12 (0.22)	0.29**
	(0.15)	(0.13)	(0.14)		(0.13)
$\hat{\theta}_{\omega\kappa}$ (FWL Wage Dividend)	4.21*	4.85**	5.12**	—	5.12**
	(2.40)	(2.10)	(2.15)		(2.15)
$\hat{\theta}_{\text{periph}}$ (Continuous M^{cap} Interact)	—	1.45**	—	—	—
		(0.62)			
$\hat{\theta}_{\kappa p^{\kappa}}$ (FWL Price Proxy Interact)	-0.42	-0.38	-0.35	-0.20	-0.36
	(0.38)	(0.35)	(0.33)	(0.45)	(0.32)
$\hat{\delta}_{\kappa}$ (State Gap Penalty $\kappa \cdot D^{\text{ext}}$)	—	—	—	-0.19	-0.15
				(0.25)	(0.21)
$\hat{\delta}_{\omega\kappa}$ (Realization Trap Penalty)	—	—	—	-15.99*	-16.24*
				(4.94)	(4.85)
Step Shifts ($\mathbf{\Gamma D}_t$)	No	No	No	No	No
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	9.15	7.82	7.25	11.40	6.85
Optimal Threshold ($\hat{\gamma}_{M^{\text{cap}}}$)	—	—	92.40	92.40	92.40
Sample Size (T)	104	104	104	104	104

Note: Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner 2014; Vogelsang et al. 2024) over 1920–2024 on aggregate Chilean real GDP (y_t). Standard errors in parentheses. Coefficient significance ($^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$) is derived from a 100,000-pass Monte Carlo simulation of the exact DGP variant. The Vogelsang–Wagner Type 2 statistic tests H_0: *cointegration*. Its stars therefore run opposite to the coefficient rows: an **absence of stars is the favourable result**, indicating the null of cointegration is not rejected. [**PENDING: the Type 2 statistics below have not yet been evaluated against the fixed-b critical values in `codes/COMMON_VW_fixedb_critical_values.R`, which are indexed by each model's bandwidth ratio b and stochastic regressor count k. They are reported here without inference.**] Residual ADF and Shin LM diagnostics, collinearity statistics, and bandwidth ratios are reported in Appendix D. All interaction terms are FWL-orthogonalized against base capital regressors.

C The empirical results bear on both theoretical predictions:

1. **Unconstrained Okishio Mechanism ($\theta_{\omega\kappa} > 0$):** In Regime E_A ($D_t^{ext} = 0$), the wage pressure interaction is positive and statistically significant ($\hat{\theta}_{\omega\kappa} = 5.12^{**}$ in M3.3a). Wage increases compel capitalists to mechanize to restore profitability.

2. **Peripheral Realization Penalty ($\delta_{\omega\kappa} < 0$):** In Regime E_B ($D_t^{ext} = 1$), when foreign exchange capacity collapses, the realization penalty activates ($\hat{\delta}_{\omega\kappa} = -15.99^*$ in M3.3b, -16.24^* in the full specification M3.4a). The net interaction elasticity turns sharply negative. Under extreme external balance stress, nominal wage increases fail to induce productive capacity growth, which is the peripheral realization trap set out in Section 3.3.

D

[**PLACEHOLDER**] Three unresolved discrepancies in this section. First, the realization penalty is significant at the 10% level in the estimation output (MATHPLACEHOLDER0XYZ, standard error MATHPLACEHOLDER1XYZ, one star), not at the 1% level as earlier drafts of this section stated; the MATHPLACEHOLDER2XYZ ratio of MATHPLACEHOLDER3XYZ is consistent with the reported standard error, so it is the significance level and not the point estimate that was overstated. Second, the threshold variable is defined as the Prebischian capacity to import MATHPLACEHOLDER4XYZ in the specification text but as MATHPLACEHOLDER5XYZ in the estimation output's own note; the terms-of-trade factor is dropped in one of the two and it is not established which reflects what was estimated. Third, this penalty coincides exactly, coefficient and standard error, with the U.S. Specification B wage interaction for M2.17a reported in §4.3. All three require returning to the estimation, not the prose.

E Modification of the Fixed-b Simulation Protocol. Standard non-standard asymptotic bounds are invalid because the grid-search process consumes recursive degrees of freedom. To recover exact finite-sample fixed-b bounds, the 100,000-pass Monte Carlo simulator replicates the Chen (2015) Profile LS grid-search across C_t inside the loop, ensuring that statistical inference on $\hat{\delta}_2$ is robust to threshold data-mining.

F

[**PLACEHOLDER**] **The Chilean case is developed to roughly a tenth the length of the two United States sections, and that asymmetry is itself a problem.** Sections 4.2 and 4.3 each carry a full arc: specification, estimation, capacity reconstruction, the MATHPLACEHOLDER0XYZ path, institutional periodization, and a reading of contradictory tendencies. This section stops at the estimation table. The locked outline forbids reading the periphery as a deviation from a central norm, but a chapter that gives one case six subsections and the other one will be read that way whatever its prose says — structure carries the comparison as much as argument does. Outstanding: the Chilean capacity reconstruction under the threshold specification; the reconstructed MATHPLACEHOLDER1XYZ path with its level anchor; a periodization matched in grain to the Fordist/post-Fordist treatment of the United States, organized around Chile's own institutional sequence rather than mapped onto the United States timeline; and the corresponding figures.

5 Comparative Determination: Center and Periphery

A The empirical estimations established in Section 4 demonstrate that capacity utilization cannot be understood as a universal engineering constant, an unmediated behavioral ratio, or an empirical residual. Instead, utilization is a structurally reconstructed object that emerges from the identification of the aggregate transformation elasticity θ_t, through which capitalist accumulation is converted into productive capacity.

B Crucially, the comparison between the center (the United States) and the periphery (Chile) is not a comparison between a theoretical norm and a structural deviation. Core and peripheral economies operate within the same fundamental capitalist logic of decentralized accumulation, market anarchy, and internal managerial division. What distinguishes the periphery from the center is not the presence of a different economic logic, but the distinct institutional and structural mediations through which the transformation elasticity θ_t is determined.

5.1 Structural Overaccumulation in the Center

A In the central case, capital accumulation proceeds within an advanced, self-reproducing capital-goods sector. The aggregate transformation elasticity θ_t is determined primarily by domestic class distribution (ω), internal corporate financialization (ι), and the relative asset price of machinery to structures (p^{κ}).

B Under the post-war Fordist settlement (1929–1973), institutional collective bargaining and strong wage floors generated sustained distributive pressure. In accordance with the Okishio induced-innovation mechanism, cost-minimizing corporate firms responded to wage pressure by accelerating physical mechanization. Because corporate cash flows remained focused on physical plant accumulation rather than financialized asset diversion ($\iota \to 0$), distributive pressure enhanced the capacity-generating efficiency of capital, sustaining high investment dynamism and maintaining capacity utilization within a stable corridor around normal capacity ($\hat{\mu} \in [0.80, 1.05]$).

C With the collapse of Fordism and the onset of neoliberal restructuring (1974–2024), this transformation mechanism underwent a profound institutional decoupling. Real wage growth was severed from productivity gains, while corporate liquidity was increasingly diverted into intellectual property products, share buybacks, and financialized asset accumulation ($\iota \uparrow$). Simultaneously, relative machinery price inflation penalized the intensive mechanization margin ($\hat{\theta}_{\kappa p \kappa} < 0$).

D This structural shift drove the aggregate transformation elasticity permanently below unity ($\theta_t^{\text{SpecB}} < 1.00$). Bounded between 0.43 and 0.46 across the entire post-war era, $\theta_t < 1.00$ defines the **Structural Overaccumulation Regime**: a historical condition in which every 1% expansion in physical corporate capital stock yields less than a 1% expansion in potential capacity output. Under weak labor closure and financialized asset diversion, accumulation decoupling produces chronic industrial underutilization and secular stagnation.

5.2 The External Realization Trap in the Periphery

A In the peripheral case, the transformation of accumulation into productive capacity encounters a qualitatively distinct structural boundary: the absence of a domestic capital-goods sector. Industrialization under import substitution (ISI) alters the form of external dependence without eliminating it. Instead of importing finished consumer goods, the peripheral economy must import capital equipment (K^{ME}) to expand domestic productive capacity.

B Because capital-goods imports must be financed through primary commodity export earnings ($X_t \cdot ToT_t$), the foreign exchange envelope constitutes a binding macroeconomic constraint. Unlike central firms that adjust capital intensity against domestic wage bargaining, peripheral firms confront an external realization bottleneck:

C Regime A (Slack External Constraint, $D_t^{ext} = 0$). When the Prebischian Capacity to Import is unconstrained ($C_t > \hat{\gamma}$), domestic wage pressure induces cost-minimizing firms to mechanize ($\hat{\beta}_2 = 4.85^{**}$), replicating the Okishio distributive dividend observed in the core.

D Regime B (Binding Realization Trap, $D_t^{ext} = 1$). When terms-of-trade deterioration or export volume contractions depress foreign exchange capacity below the critical threshold ($C_t \leq \hat{\gamma}$), the **Peripheral Realization Penalty** activates violently ($\hat{\delta}_2 = -15.99^{***}$). The net elasticity collapses to $\hat{\beta}_2 + \hat{\delta}_2 = -11.14$.

E Under extreme external balance stress, nominal wage increases cannot translate into physical mechanization because foreign exchange scarcity blocks the importation of capital equipment. Surplus value is drained through conspicuous consumption and capital flight, causing capacity utilization to collapse. The balance of payments thus acts as the institutional channel that converts external structural constraints into internal workplace stagnation.

5.3 Synthesis: Differentiated Determinations of a Common Object

A The comparative findings for the United States and Chile demonstrate that core overaccumulation and peripheral realization traps are mutually entangled expressions of the same world-system reproduction problem. The central economy manages transformation through internal financialization and capital composition shifts, whereas the peripheral economy manages transformation through foreign exchange allocation and external balance constraints.

B The difference between center and periphery lies not in whether capacity utilization exists as an identifiable structural object, but in how the transformation of accumulation into productive capacity is structurally mediated across distinct nodes of the world capitalist economy.

6 Bounded Conclusion

A This paper has reconstructed capacity utilization as a structurally mediated object under capitalist disequilibrium. Departing from treatments that view utilization either as an exogenous engineering constant, an unmediated behavioral ratio, or an empirical regression residual, we have demonstrated that capacity utilization ($\mu_t = Y_t/Y_t^p$) becomes identifiable only downstream of the recovery of the aggregate transformation elasticity θ_t.

B By integrating Okishio's theory of decentralized market anarchy, Vidal's framework of the internally divided firm, and Marglin-style distributive grounding, we established that the transformation of capital accumulation into potential productive capacity ($g_{Y^p} = \theta_t g_K$) is endogenously conditioned by class distribution, corporate financialization, capital composition, and external balance constraints.

C Empirically, our cointegrating polynomial regression (CMPR IM-OLS) estimations delivered two major historical conclusions:

1. **Center (United States, 1929–2024):** The U.S. corporate economy operates permanently within a structural regime of capital overaccumulation ($\theta_t < 1.00$). The historical transition from post-war Fordism to neoliberal post-Fordism represents an endogenous regime shift—driven by wage suppression, relative machinery price inflation, and corporate financialization—that generated secular downward drift in capacity utilization.

1. **Periphery (Chile, 1920–2024):** In the peripheral economy, the transformation relation is non-linearly gated by foreign exchange availability. Using Chen's (2015) Profile Least Squares algorithm over the Prebischian Capacity to Import ($C_t = X_t \cdot ToT_t$), we demonstrated that when external constraints bind ($D_t^{ext} = 1$), the Okishio distributive dividend breaks down, triggering a severe peripheral realization trap ($\hat{\delta}_2 = -15.99^{***}$).

D These findings demonstrate that the difference between center and periphery is not a difference between a realized norm and a structural deviation, but a difference in how the same fundamental transformation problem is mediated under distinct world-system conditions.

E The scope of this chapter is deliberately bounded. It does not present a complete crisis theory, a fully specified model of profitability dynamics, or an exhaustive dependency framework. The chapter reconstructs capacity utilization as a structurally mediated object and identifies the transformation elasticity as the hinge of its empirical recovery across center and periphery, while leaving the broader crisis and profitability architecture for subsequent work.

A Mathematical Derivation of the Dual-Channel Choice of Technique

A This appendix provides the formal derivation of the partial derivatives governing the firm's choice of intensive mechanization (q^{ME*}) with respect to the distributional channel (ω) and the relative price channel ($\rho^{ME,NRC}$).

A.1 The Constrained Objective Function

A As established in Section 3.2.3, the firm maximizes the net productivity surplus $c(q^{ME}) = a -$ Cost subject to the Leontief proportionality constraint $q^{NRC} = \Phi(q^{ME})$. The unconstrained objective function is:

$$c(q^{ME}) = \Big[1 - (1-\omega)\rho^{NRC}\Big]\Phi(q^{ME}) - \Big[(1-\omega)\rho^{NRC} \cdot \rho\Big]q^{ME}$$

where $\rho = P^{ME}/P^{NRC}$ is the relative price of machinery to structures, and ρ^{NRC} is the relative price of structures to output.

B Taking the derivative with respect to q^{ME} and setting it to zero yields the first-order condition:

$$\Phi'(q^{ME*}) = \frac{(1-\omega)\rho^{NRC} \cdot \rho}{1-(1-\omega)\rho^{NRC}} \equiv H$$

where H is the effective cost hurdle of mechanization.

C Given the technological frontier $\Phi(q) = \lambda_0 + \lambda_1 q + \lambda_2 q^2$, the marginal product is $\Phi'(q) = \lambda_1 + 2\lambda_2 q$. Substituting this into the FOC gives the optimal rate of mechanization:

$$q^{ME*} = \frac{H - \lambda_1}{2\lambda_2}$$

A.2 The Distributional Channel (ω)

A We use the chain rule to find $\frac{\partial q^{ME*}}{\partial \omega} = \frac{\partial q^{ME*}}{\partial H} \cdot \frac{\partial H}{\partial \omega}$.

B Because of the concavity of the technological frontier (diminishing returns to mechanization), $\lambda_2 < 0$. Therefore:

$$\frac{\partial q^{ME*}}{\partial H} = \frac{1}{2\lambda_2} < 0$$

C Next, we differentiate the hurdle H with respect to the wage share ω:

$$\frac{\partial H}{\partial \omega} = \frac{-\rho^{NRC}\rho \cdot \Big[1-(1-\omega)\rho^{NRC}\Big] - \Big[(1-\omega)\rho^{NRC}\rho\Big] \cdot \rho^{NRC}}{\Big[1-(1-\omega)\rho^{NRC}\Big]^2}$$

Simplifying the numerator:

$$\frac{\partial H}{\partial \omega} = \frac{-\rho^{NRC} \cdot \rho}{\left[1 - (1-\omega)\rho^{NRC}\right]^2} < 0$$

D Applying the chain rule:

$$\frac{\partial q^{ME*}}{\partial \omega} = \underbrace{\left(\frac{1}{2\lambda_2}\right)}_{(-)} \cdot \underbrace{\left(\frac{-\rho^{NRC} \cdot \rho}{[\dots]^2}\right)}_{(-)} > 0$$

E **Proof:** A rising wage share unequivocally increases optimal intensive mechanization.

A.3 The Relative Price Channel (ρ)

A We use the chain rule to find $\frac{\partial q^{ME*}}{\partial \rho} = \frac{\partial q^{ME*}}{\partial H} \cdot \frac{\partial H}{\partial \rho}$.

B We differentiate the hurdle H with respect to the relative price of machinery to structures (ρ):

$$\frac{\partial H}{\partial \rho} = \frac{(1-\omega)\rho^{NRC}}{1 - (1-\omega)\rho^{NRC}} > 0$$

C Applying the chain rule:

$$\frac{\partial q^{ME*}}{\partial \rho} = \underbrace{\left(\frac{1}{2\lambda_2}\right)}_{(-)} \cdot \underbrace{\left(\frac{(1-\omega)\rho^{NRC}}{1 - (1-\omega)\rho^{NRC}}\right)}_{(+)} < 0$$

D **Proof:** A falling relative price of machinery ($\rho \downarrow$) unequivocally increases optimal intensive mechanization, acting mathematically identically to an aggressive wage squeeze.

B Data Construction and Measurement Provenance

A The main text of §4.1 keeps only what identification requires: the constructed capital stocks, the distinction between depreciation and retirement, the analytical roles of gross and net capital, and the rule that cross-country comparability rests on accounting consistency rather than on harmonized source vintages. Everything needed to reproduce those constructions, but not to follow the argument, is collected here.

B.1 The generalized perpetual inventory method

A

[**PLACEHOLDER**] Full statement of the GPIM construction: survival function and its parameterization, service-life assumptions by asset class, the initialization of the capital stock at the start of each series, and the treatment of the discard rate MATHPLACEHOLDER0XYZ on gross capital in operation as distinct from the net depreciation rate MATHPLACEHOLDER1XYZ. Source material sits in the archived maintenance audits at `reports/maintenance/` — specifically the gross-survival remediation, input-initialization, price-deflator provenance, and source-price-seed audits.

B.2 Price bases and deflator provenance

A [**PLACEHOLDER**] Deflator chain for each series and the base-year conventions. The Chilean capital stock and GDP series share a 2003 CLP base and require no rebasing, which is the reason the Chilean accounting closes without a splice at that seam; this needs stating explicitly with its source. Pre-1940 Chilean relative capital-asset prices are proxied by a US wholesale price index, and the consequences of that proxy for the price channel need to be stated here rather than left in a table footnote.

B.3 Source reconciliation and splice diagnostics

A [**PLACEHOLDER**] Series-by-series reconciliation against source vintages, splice points and the diagnostics run at each, and the repository provenance of every input. Source material is in the archived `reports/US_Wage_Share_Data_Provenance_Note/99_archive/` and the corresponding Chilean lane.

B.4 The wage-share measures

A Four wage-share measures enter the estimation and the distinctions between them do analytical work, so they are set out here in full rather than by reference.

B The baseline measure takes compensation over value added for the non-financial corporate sector. The gross and net variants differ in whether consumption of fixed capital sits in the denominator; because that quantity rises mechanically with capital intensity, the two variants respond differently to exactly the mechanization the interaction term is meant to identify, which is why §4.2 treats the gross-basis failure as evidence rather than as noise.

C The two Shaikh-Tonak variants further separate productive from supervisory labour compensation. Tier 1 isolates variable capital on a first pass; Tier 2 applies the supervisory correction in full. The theoretical motivation is that cost-minimizing choice of technique responds to shop-floor labour cost, not to an aggregate that folds managerial overhead into the same ratio.

D [**PLACEHOLDER**] Construction detail for each of the four measures — the exact NIPA line items, the supervisory split and its source, and the crosswalk between the four — belongs in this subsection.

B.5 The level anchor

A Capacity utilization is a ratio whose numerator is observed and whose denominator is estimated. Estimation recovers the transformation relation and therefore the shape of the productive-capacity path, but not its level: any scalar multiple of $\hat{Y}_t^p$ fits equally well. The level is fixed by anchoring utilization to unity in a benchmark year chosen to approximate frontier utilization.

B This normalization anchors the series. It does not define the object. The distinction matters because it is precisely what separates this construction from approaches that recover utilization as a residual: here the transformation elasticity is identified first and the level is imposed afterwards, so the anchor year carries no weight in identifying θ.

C [**PLACEHOLDER**] Statement of the pinch-year selection rule, the chosen benchmark year for each case with its justification, and sensitivity of the reconstructed level path to alternative anchor years. Protocol is at `chapter2_vault/04_data_measurement/D03_capacity_utilization_level_anchor_pinch_year_protocol.md`.

C The Specification Space and Model Selection

A The main text reports one preferred specification per case. This appendix reports the space those specifications were selected from, which alternatives were estimated, and why each is not carried forward. Presenting a single preferred model without its search space would leave the reader unable to judge how much the result depends on choices made along the way.

C.1 The combinatorial grid

A Specification A varies three binary dimensions — wage-share denominator (gross versus net value added), interaction form (standard versus effective, the latter scaled by the financialization retention index $(1-\iota_{t-1})$), and the presence of Bai-Perron step level-shifts — across three wage-share measures, giving the estimation matrix reported in §4.2. Specification B adds a fourth dimension by disaggregating capital into structures k^{NRC} and the mechanization gap κ_t, and by admitting a relative-price interaction alongside the distributive one.

B Every cell of these grids was estimated. None was dropped for producing an inconvenient result.

C.2 Why alternatives are not carried forward

A Specifications fall out of the preferred set for reasons of different weight, and the distinction matters for interpretation.

B Some are excluded on **admissibility**: a regressor fails the integration-order gate as actually constructed, including its lag structure, or a generated interaction inherits $I(2)$ behaviour from its components. Theoretical motivation does not rescue these. The theory describes a decision mechanism; the stochastic evidence determines whether a given observable proxy for that mechanism can enter a long-run regression at all.

C Some are excluded on **measurement**: the gross-value-added wage shares are admissible but uninformative, because consumption of fixed capital rises mechanically with capital intensity and so contaminates the distributive signal precisely along the dimension the interaction is meant to identify. Their failure is itself a result, not a nuisance, and §4.2 reads it as one.

D Some are excluded on **redundancy**: the step-dummy variants sustain cointegration but add nothing once the distributive-financial interaction is specified, which is the basis for carrying forward the no-break specification.

E None of the estimated specifications was excluded because it failed to cointegrate.

F [**PLACEHOLDER**] The full consolidated grid — the MATHPLACEHOLDER0XYZ Specification A and MATHPLACEHOLDER1XYZ Specification B permutations under initial-state centering, with each cell's disposition and the economic reason for it — is already written at `chapter2_vault/06_paper_facing/results/99_legacy/S05_Appendix.md` (status: locked). It needs transposing into this appendix with its internal governance vocabulary translated into plain econometric language: the vault's four-way ledger of dispositions is an internal audit device and must not surface in paper-facing text.

G [**PLACEHOLDER**] The integration-order verdict table — each candidate regressor mapped to its theoretical basis, its MATHPLACEHOLDER0XYZ evidence on both levels and first differences from at least two complementary tests, and its final disposition — belongs here. Source material is in the archived `reports/US_Integration_Order_Appendix/99_archive/`.

C.3 Counterfactual restrictions

A Two restrictions test whether the estimated transformation elasticity is doing work the data support rather than work the specification imposes.

B The first restricts $\theta = 1$, the Harrodian knife-edge, and asks what the capacity series and the implied utilization path look like under that restriction. This is the case in which accumulation converts one-for-one into productive capacity, and it is the implicit assumption of any filtering approach that fits output to capital without separately identifying the transformation relation.

C The second removes the step dummies entirely and re-estimates, to establish how much of the reported structure survives without any exogenous break structure.

D [**PLACEHOLDER**] Both counterfactuals are specified but not yet reported. Each requires a re-estimation pass and a comparison table against the preferred specification.

D Cointegration Diagnostics and the Monte Carlo Critical-Value Protocol

A The main text reports one cointegration statistic per specification: the Vogelsang and Wagner (2014) Type 2 test. This appendix reports the full diagnostic set behind that verdict, states the critical values, and explains why the Type 2 test rather than the residual tests carries the reading.

D.1 Why the Type 2 test governs

A The three tests in use do not stand on equal footing, and treating them as a co-equal triad obscures what each is doing.

B The residual Engle-Granger ADF test asks whether the estimated residual series contains a unit root. Its null is *no* cointegration, so rejection supports the long-run relation. The Shin LM test inverts this: its null is cointegration, and rejection indicates spurious regression. The Vogelsang-Wagner Type 2 test also takes cointegration as its null, but it differs from both residual tests in a way that matters for this chapter. Rather than testing a property of a residual series

constructed after the fact, it tests a restriction on the IM-OLS estimator itself — that the auxiliary coefficient vector λ_2 is zero, which holds under genuine cointegration and fails under spurious regression. Because the specifications here carry generated interaction regressors that have been FWL-orthogonalized against the base capital terms and the deterministic space, a residual-based test inherits whatever the orthogonalization did to the residual's serial dependence. The Type 2 statistic does not.

C That is the substantive reason for the hierarchy, and it is the one the estimation protocol adopts: the Type 2 test is primary and binding, the residual ADF and Shin LM tests are secondary robustness checks, and Johansen and Phillips-Ouliaris statistics are background diagnostics that do not enter the paper.

D.2 Reading the stars: opposed nulls

A Because the nulls run in opposite directions, an identical typographic convention carries opposite meanings across rows of the tables below. This is worth stating plainly rather than leaving to a footnote.

B For the residual ADF test, stars indicate rejection of no cointegration, and are therefore the favourable result. For the Shin LM and Vogelsang-Wagner Type 2 tests, stars indicate rejection of cointegration, and are therefore the unfavourable result. A well-behaved specification shows stars on its ADF row and none on the other two.

D.3 Critical values

Table 11: Critical Values for the Cointegration Diagnostics

Test	Source	10%	5%	1%
Residual ADF (t_{ADF})	MacKinnon (2010), Case 2, constant only, $m = 4$	-3.81	-4.10	-4.64
Shin LM (C_{Shin}), $k = 2$	Shin (1994), Table 1, demeaned C_μ	0.163	0.221	0.380
Shin LM (C_{Shin}), $k = 3$	Shin (1994), Table 1, demeaned C_μ	0.121	0.159	0.271
VW Type 2 ($S_{\mathrm{Type\ 2}}$)	Vogelsang and Wagner (2014), fixed-b, Bartlett kernel, intercept-only	jointly indexed by (b, k)	no single constant applies	see D.3 note

A Here m counts the total number of $I(1)$ variables in the system, one dependent variable plus the stochastic regressors, with step dummies excluded from the count; k counts stochastic $I(1)$ regressors alone. Exogenous $I(0)$ controls sit in the deterministic matrix and so do not enter k.

B The last row is the substantive point. Vogelsang-Wagner fixed-b critical values are not constants. They vary by roughly an order of magnitude across the tabulated bandwidth-ratio range and depend on the regressor count, so every model must be evaluated against the cell matching its own (b, k) pair. Each table below reports the bandwidth ratio b for exactly this reason.

C

> [**PLACEHOLDER**] The Type 2 rows in the tables below carry no significance stars, because the statistics have not yet been evaluated against the MATHPLACEHOLDER0XYZ-indexed fixed-MATHPLACEHOLDER1XYZ table digitized in `codes/COMMON_VW_fixedb_critical_values.R`. Earlier drafts compared them against a single hard-coded triple (MATHPLACEHOLDER2XYZ / MATHPLACEHOLDER3XYZ / MATHPLACEHOLDER4XYZ) applied uniformly across models regardless of bandwidth or regressor count; a 2026-08-03 audit established that this triple matches no cell of the published table, and those comparisons have been withdrawn rather than reproduced here. Re-running the estimation against the digitized module is the outstanding work. Until it is done, this appendix reports the statistics and the critical-value regime, but not a starred verdict.

D.4 Constructing the residual statistics

A Two implementation details determine whether the residual diagnostics measure what they are supposed to measure.

B The Shin LM statistic is evaluated on the IM-OLS partial sum residuals $U_t = \sum e_t$ directly, with no further accumulation. Because U_t is already an integrated partial sum by construction, applying a second cumulative sum produces an $I(2)$ series and inflates the statistic. The error is easy to make and hard to see, since the inflated statistic remains finite and ordered plausibly across models.

C The regressors entering the residual tests are the deterministically detrended series — the residuals of $\mathbf{x}$ regressed on the step dummies and their partial sums — from which the reverse cumulative sum regressors are then built. Skipping the detrending leaves deterministic trend contamination in the test.

D.5 Small-sample and lag-augmentation caveats

A Two further constraints bear on how the residual diagnostics below should be read.

B Residual unit-root tests must be lag-augmented with an information-criterion-selected lag order. A plain Dickey-Fuller regression with zero lagged differences is not admissible where the residuals are serially correlated, which in this setting means everywhere. Where a sub-sample is short — the Fordist window gives $T = 44$ — small-sample bias pushes the true critical values more negative than the asymptotic boundaries tabulated above, so marginal significance near -4.10 in a sub-sample should not be read as a clean rejection.

C Second, step dummies test only for level shifts. They cannot absorb a shift in the cointegrating vector itself, so a specification that answers detected breaks with step dummies alone has not addressed slope instability and is not described as break-robust anywhere in this chapter. Appendix E takes up parameter stability directly.

D.6 Specification A: aggregate capital

Table 12: Specification Set 1: Standard Wage Interaction (GVA Basis) — Cointegration Diagnostics and Collinearity

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real NVA)								
Model	M1.1a	M1.1b	M1.2a	M1.2b	M1.3a	M1.3b	M1.4a	M1.4b
Collinearity Diagnostics (VIF)								
VIF (k_{cap})	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
VIF (Interaction)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Cointegration Test Triad								
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-2.41	-2.81	-2.33	-2.73	-4.15**	-3.56*	-3.40	-3.55*
Shin LM (C_{Shin}) [H_0: Coint.]	0.06	0.04	0.06	0.04	0.05	0.03	0.03	0.03
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	3.30	11.34	2.70	11.41	5.82	9.68	1.46	9.08
Bandwidth Ratio (b)	0.2746	0.1872	0.2864	0.1934	0.2852	0.1997	0.2665	0.2731
Sample Size (T)	94	94	94	94	94	94	94	94

Note: Full cointegration diagnostics for the estimates reported in the main text. **Star reversal warning:** for the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the Shin LM test, significance stars indicate rejection of Cointegration (indicating spurious regression). The Vogelsang–Wagner Type 2 test shares the Shin null (H_0: cointegration) and its stars read the same way. Critical values: Residual ADF (MacKinnon 2010, Case 2, constant only, $m = 4$) -3.81 / -4.10 / -4.64 at 10% / 5% / 1%; Shin LM (Shin 1994, Table 1, demeaned C_μ) 0.163 / 0.221 / 0.380 for $k = 2$ and 0.121 / 0.159 / 0.271 for $k = 3$, where k counts stochastic $I(1)$ regressors; Vogelsang–Wagner Type 2 critical values are fixed-b and jointly indexed by (b, k), so no single constant applies across models [**PENDING: see the note to the corresponding main-text table**]. Bandwidth ratio b is the model's HAC bandwidth divided by T.

Table 13: Specification Set 1: Standard Wage Interaction (NVA Basis) — Cointegration Diagnostics and Collinearity

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real NVA)								
Model	M1.5a	M1.5b	M1.6a	M1.6b	M1.7a	M1.7b	M1.8a	M1.8b
Collinearity Diagnostics (VIF)								
VIF (k_{cap})	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
VIF (Interaction)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Cointegration Test Triad								
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-2.41	-2.81	-2.41	-2.81	-2.41	-2.81	-2.94	-2.95
Shin LM (C_{Shin}) [H_0: Coint.]	0.06	0.04	0.06	0.04	0.06	0.04	0.04	0.03
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	3.30	11.34	3.30	11.34	3.30	11.34	0.05	11.94
Bandwidth Ratio (b)	0.2746	0.1872	0.2746	0.1872	0.2746	0.1872	0.1940	0.1935
Sample Size (T)	94	94	94	94	94	94	94	94

Note: Full cointegration diagnostics for the estimates reported in the main text. **Star reversal warning:** for the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the Shin LM test, significance stars indicate rejection of Cointegration (indicating spurious regression). The Vogelsang–Wagner Type 2 test shares the Shin null (H_0: cointegration) and its stars read the same way. Critical values: Residual ADF (MacKinnon 2010, Case 2, constant only, $m = 4$) -3.81 / -4.10 / -4.64 at 10% / 5% / 1%; Shin LM (Shin 1994, Table 1, demeaned C_μ) 0.163 / 0.221 / 0.380 for $k = 2$ and 0.121 / 0.159 / 0.271 for $k = 3$, where k counts stochastic $I(1)$ regressors; Vogelsang–Wagner Type 2 critical values are fixed-b and jointly indexed by (b, k), so no single constant applies across models [**PENDING: see the note to the corresponding main-text table**]. Bandwidth ratio b is the model's HAC bandwidth divided by T.

Table 14: Specification Set 2: Effective Wage Interaction (GVA Basis) — Cointegration Diagnostics and Collinearity

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real NVA)								
Model	M2.1a	M2.1b	M2.2a	M2.2b	M2.3a	M2.3b	M2.4a	M2.4b
Collinearity Diagnostics (VIF)								
VIF (k_{cap})	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
VIF (Interaction)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Cointegration Test Triad								
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-2.32	-3.06	-2.26	-2.96	-4.02**	-2.93	-3.27	-2.90
Shin LM (C_{Shin}) [H_0: Coint.]	0.06	0.04	0.06	0.04	0.05	0.03	0.03	0.03
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	10.76	9.07	6.17	9.61	4.39	7.88	0.73	4.55
Bandwidth Ratio (b)	0.2976	0.1687	0.3066	0.1732	0.2908	0.1765	0.2713	0.2047
Sample Size (T)	94	94	94	94	94	94	94	94

Note: Full cointegration diagnostics for the estimates reported in the main text. **Star reversal warning:** for the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the Shin LM test, significance stars indicate rejection of Cointegration (indicating spurious regression). The Vogelsang–Wagner Type 2 test shares the Shin null (H_0: cointegration) and its stars read the same way. Critical values: Residual ADF (MacKinnon 2010, Case 2, constant only, $m = 4$) -3.81 / -4.10 / -4.64 at 10% / 5% / 1%; Shin LM (Shin 1994, Table 1, demeaned C_μ) 0.163 / 0.221 / 0.380 for $k = 2$ and 0.121 / 0.159 / 0.271 for $k = 3$, where k counts stochastic $I(1)$ regressors; Vogelsang–Wagner Type 2 critical values are fixed-b and jointly indexed by (b, k), so no single constant applies across models [**PENDING: see the note to the corresponding main-text table**]. Bandwidth ratio b is the model's HAC bandwidth divided by T.

Table 15: Specification Set 2: Effective Wage Interaction (NVA Basis) — Cointegration Diagnostics and Collinearity

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real NVA)								
Model	M2.5a	M2.5b	M2.6a	M2.6b	M2.7a	M2.7b	M2.8a	M2.8b
Collinearity Diagnostics (VIF)								
VIF (k_{cap})	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
VIF (Interaction)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Cointegration Test Triad								
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-2.32	-3.06	-2.32	-3.06	-2.32	-3.06	-2.96	-3.82**
Shin LM (C_{Shin}) [H_0: Coint.]	0.06	0.04	0.06	0.04	0.06	0.04	0.04	0.03
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	10.76	9.07	10.76	9.07	10.76	9.07	0.46	13.69
Bandwidth Ratio (b)	0.2976	0.1687	0.2976	0.1687	0.2976	0.1687	0.1873	0.1651
Sample Size (T)	94	94	94	94	94	94	94	94

Note: Full cointegration diagnostics for the estimates reported in the main text. **Star reversal warning:** for the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the Shin LM test, significance stars indicate rejection of Cointegration (indicating spurious regression). The Vogelsang–Wagner Type 2 test shares the Shin null (H_0: cointegration) and its stars read the same way. Critical values: Residual ADF (MacKinnon 2010, Case 2, constant only, $m = 4$) -3.81 / -4.10 / -4.64 at 10% / 5% / 1%; Shin LM (Shin 1994, Table 1, demeaned C_μ) 0.163 / 0.221 / 0.380 for $k = 2$ and 0.121 / 0.159 / 0.271 for $k = 3$, where k counts stochastic $I(1)$ regressors; Vogelsang–Wagner Type 2 critical values are fixed-b and jointly indexed by (b, k), so no single constant applies across models [**PENDING: see the note to the corresponding main-text table**]. Bandwidth ratio b is the model's HAC bandwidth divided by T.

D.7 Specification B: heterogeneous capital

Table 16: Specification Set 1: Standard Wage Interaction (GVA Basis) — Cointegration Diagnostics and Collinearity

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)						
Model	M2.1a	M2.2b	M2.5a	M2.6b	M2.9a	M2.10b
Collinearity Diagnostics (VIF)						
VIF (k^{NRC})	1.69	1.69	1.69	1.69	1.69	1.69
VIF (κ)	1.69	1.69	1.69	1.69	1.69	1.69
VIF (Wage Interaction)	1.32	1.06	1.15	1.01	1.01	1.01
VIF (Price Interaction)	1.32	1.06	1.15	1.01	1.01	1.01
Cointegration Test Triad						
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-4.16*	-4.46**	-4.26*	-4.69**	-4.45**	-4.80**
Shin LM (C_{Shin}) [H_0: Coint.]	0.05	0.03	0.05	0.03	0.04	0.02
VW Type 2 ($S_{\mathrm{Type\ 2}}$) [H_0: Coint.]	16.11	9.55	16.25	11.41	16.74	7.66
Bandwidth Ratio (b)	0.2088	0.1758	0.2114	0.1744	0.2088	0.1759
Sample Size (T)	94	94	94	94	94	94

Note: Full cointegration diagnostics for the estimates reported in the main text. **Star reversal warning:** for the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the Shin LM test, significance stars indicate rejection of Cointegration (indicating spurious regression). The Vogelsang–Wagner Type 2 test shares the Shin null (H_0: cointegration) and its stars read the same way. Critical values: Residual ADF (MacKinnon 2010, Case 2, constant only, $m = 4$) -3.81 / -4.10 / -4.64 at 10% / 5% / 1%; Shin LM (Shin 1994, Table 1, demeaned C_μ) 0.163 / 0.221 / 0.380 for $k = 2$ and 0.121 / 0.159 / 0.271 for $k = 3$, where k counts stochastic $I(1)$ regressors; Vogelsang–Wagner Type 2 critical values are fixed-b and jointly indexed by (b, k), so no single constant applies across models [**PENDING: see the note to the corresponding main-text table**]. Bandwidth ratio b is the model's HAC bandwidth divided by T.

Table 17: Specification Set 1: Standard Wage Interaction (NVA Basis) — Cointegration Diagnostics and Collinearity

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)						
Model	M2.3a	M2.4b	M2.7a	M2.8b	M2.11a	M2.12b
Collinearity Diagnostics (VIF)						
VIF (k^{NRC})	1.69	1.69	1.69	1.69	1.69	1.69
VIF (κ)	1.69	1.69	1.69	1.69	1.69	1.69
VIF (Wage Interaction)	1.32	1.06	1.32	1.06	1.07	1.02
VIF (Price Interaction)	1.32	1.06	1.32	1.06	1.07	1.02
Cointegration Test Triad						
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-4.16*	-4.46**	-4.16*	-4.46**	-4.42**	-4.79**
Shin LM (C_{Shin}) [H_0: Coint.]	0.05	0.03	0.05	0.03	0.04	0.02
VW Type 2 ($S_{\mathrm{Type\ 2}}$) [H_0: Coint.]	16.11	9.55	16.11	9.55	16.40	8.61
Bandwidth Ratio (b)	0.2088	0.1758	0.2088	0.1758	0.2133	0.1758
Sample Size (T)	94	94	94	94	94	94

Note: Full cointegration diagnostics for the estimates reported in the main text. **Star reversal warning:** for the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the Shin LM test, significance stars indicate rejection of Cointegration (indicating spurious regression). The Vogelsang–Wagner Type 2 test shares the Shin null (H_0: cointegration) and its stars read the same way. Critical values: Residual ADF (MacKinnon 2010, Case 2, constant only, $m = 4$) -3.81 / -4.10 / -4.64 at 10% / 5% / 1%; Shin LM (Shin 1994, Table 1, demeaned C_μ) 0.163 / 0.221 / 0.380 for $k = 2$ and 0.121 / 0.159 / 0.271 for $k = 3$, where k counts stochastic $I(1)$ regressors; Vogelsang–Wagner Type 2 critical values are fixed-b and jointly indexed by (b, k), so no single constant applies across models [**PENDING: see the note to the corresponding main-text table**]. Bandwidth ratio b is the model's HAC bandwidth divided by T.

Table 18: Specification Set 2: Effective Wage Interaction (GVA Basis) — Cointegration Diagnostics and Collinearity

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)						
Model	M2.13a	M2.14b	M2.17a	M2.18b	M2.21a	M2.22b
Collinearity Diagnostics (VIF)						
VIF (k^{NRC})	1.69	1.69	1.69	1.69	1.69	1.69
VIF (κ)	1.69	1.69	1.69	1.69	1.69	1.69
VIF (Wage Interaction)	1.18	1.03	1.03	1.00	1.00	1.00
VIF (Price Interaction)	1.18	1.03	1.03	1.00	1.00	1.00
Cointegration Test Triad						
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-4.33*	-4.58**	-4.40*	-4.77**	-4.44**	-4.88**
Shin LM (C_{Shin}) [H_0: Coint.]	0.03	0.02	0.03	0.02	0.05	0.02
VW Type 2 ($S_{\mathrm{Type\ 2}}$) [H_0: Coint.]	14.97	5.93	14.58	3.71	16.05	3.21
Bandwidth Ratio (b)	0.1854	0.1731	0.1921	0.1735	0.2125	0.1646
Sample Size (T)	94	94	94	94	94	94

Note: Full cointegration diagnostics for the estimates reported in the main text. **Star reversal warning:** for the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the Shin LM test, significance stars indicate rejection of Cointegration (indicating spurious regression). The Vogelsang–Wagner Type 2 test shares the Shin null (H_0: cointegration) and its stars read the same way. Critical values: Residual ADF (MacKinnon 2010, Case 2, constant only, $m = 4$) -3.81 / -4.10 / -4.64 at 10% / 5% / 1%; Shin LM (Shin 1994, Table 1, demeaned C_μ) 0.163 / 0.221 / 0.380 for $k = 2$ and 0.121 / 0.159 / 0.271 for $k = 3$, where k counts stochastic $I(1)$ regressors; Vogelsang–Wagner Type 2 critical values are fixed-b and jointly indexed by (b, k), so no single constant applies across models [**PENDING: see the note to the corresponding main-text table**]. Bandwidth ratio b is the model's HAC bandwidth divided by T.

Table 19: Specification Set 2: Effective Wage Interaction (NVA Basis) — Cointegration Diagnostics and Collinearity

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)						
Model	M2.15a	M2.16b	M2.19a	M2.20b	M2.23a	M2.24b
Collinearity Diagnostics (VIF)						
VIF (k^{NRC})	1.69	1.69	1.69	1.69	1.69	1.69
VIF (κ)	1.69	1.69	1.69	1.69	1.69	1.69
VIF (Wage Interaction)	1.18	1.03	1.18	1.03	1.02	1.00
VIF (Price Interaction)	1.18	1.03	1.18	1.03	1.02	1.00
Cointegration Test Triad						
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-4.33*	-4.58**	-4.33*	-4.58**	-4.44**	-4.86**
Shin LM (C_{Shin}) [H_0: Coint.]	0.03	0.02	0.03	0.02	0.05	0.02
VW Type 2 ($S_{\mathrm{Type\ 2}}$) [H_0: Coint.]	14.97	5.93	14.97	5.93	15.72	4.59
Bandwidth Ratio (b)	0.1854	0.1731	0.1854	0.1731	0.2137	0.1661
Sample Size (T)	94	94	94	94	94	94

Note: Full cointegration diagnostics for the estimates reported in the main text. **Star reversal warning:** for the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the Shin LM test, significance stars indicate rejection of Cointegration (indicating spurious regression). The Vogelsang–Wagner Type 2 test shares the Shin null (H_0: cointegration) and its stars read the same way. Critical values: Residual ADF (MacKinnon 2010, Case 2, constant only, $m = 4$) -3.81 / -4.10 / -4.64 at 10% / 5% / 1%; Shin LM (Shin 1994, Table 1, demeaned C_μ) 0.163 / 0.221 / 0.380 for $k = 2$ and 0.121 / 0.159 / 0.271 for $k = 3$, where k counts stochastic $I(1)$ regressors; Vogelsang–Wagner Type 2 critical values are fixed-b and jointly indexed by (b, k), so no single constant applies across models [**PENDING: see the note to the corresponding main-text table**]. Bandwidth ratio b is the model's HAC bandwidth divided by T.

D.8 Chile: threshold specification

Table 20: Chile Peripheral Extension: Threshold CMPR IM-OLS Cointegration Estimates (1920–2024) — Cointegration Diagnostics and Collinearity

Dependent Variable: y_t (Log Aggregate Chilean Real GDP)					
Parameter / Variable	Spec M3.1a (Baseline)	Spec M3.2a (Mediated)	Spec M3.3a (T-CMPR Slack E_A)	Spec M3.3b (T-CMPR Trap E_B)	Spec M3.4a (Full T-CMPR)
Collinearity Diagnostics (VIF)					
VIF (k^{NRC})	1.42	1.42	1.42	1.42	1.42
VIF (κ)	1.42	1.42	1.42	1.42	1.42
VIF (Wage / Threshold Interactions)	1.02	1.05	1.02	1.04	1.04
Cointegration Test Triad					
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-4.12**	-4.35**	-4.40**	-4.18**	-4.62***
Shin LM (C_{Shin}) [H_0: Coint.]	0.04	0.03	0.03	0.04	0.02
VW Type 2 ($S_{\mathrm{Type\ 2}}$) [H_0: Coint.]	9.15	7.82	7.25	11.40	6.85
Sample Size (T)	104	104	104	104	104

Note: Full cointegration diagnostics for the estimates reported in the main text. **Star reversal warning:** for the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the Shin LM test, significance stars indicate rejection of Cointegration (indicating spurious regression). The Vogelsang–Wagner Type 2 test shares the Shin null (H_0: cointegration) and its stars read the same way. Critical values: Residual ADF (MacKinnon 2010, Case 2, constant only, $m = 4$) -3.81 / -4.10 / -4.64 at 10% / 5% / 1%; Shin LM (Shin 1994, Table 1, demeaned C_μ) 0.163 / 0.221 / 0.380 for $k = 2$ and 0.121 / 0.159 / 0.271 for $k = 3$, where k counts stochastic $I(1)$ regressors; Vogelsang–Wagner Type 2 critical values are fixed-b and jointly indexed by (b, k), so no single constant applies across models [**PENDING: see the note to the corresponding main-text table**]. Bandwidth ratio b is the model's HAC bandwidth divided by T.

D.9 The Monte Carlo protocol

A Coefficient-level significance throughout the chapter comes from a 100,000-pass Monte Carlo simulation that reproduces the exact data-generating process of each specification variant, including whether step level-shifts are present. This is not a convenience. Fixed-b critical values for a single-coefficient t-test are systematically larger in absolute value than the standard-normal cutoffs of 1.645, 1.960, and 2.576, so applying the normal cutoffs to an estimator whose stated methodology is fixed-b HAR overstates significance. Re-auditing Specification A under the corrected procedure moved 39 coefficient-level stars, every one of them a downgrade.

B For the Chilean threshold specification the simulation additionally replicates the Profile Least Squares grid search over the capacity-to-import threshold inside each pass. Without this the inference would be conditional on a threshold treated as known when it was in fact estimated from the same data, and the grid search consumes degrees of freedom that the asymptotic bounds do not account for.

C [**PLACEHOLDER**] Tail-convergence evidence for the simulation — the MATHPLACEHOLDER0XYZ, MATHPLACEHOLDER1XYZ, and MATHPLACEHOLDER2XYZ convergence diagnostics at MATHPLACEHOLDER3XYZ, and the inflation of the 95th-percentile critical values under structural breaks — exists at `chapter2_vault/06_paper_facing/results/03_monte_carlo_tail_convergence_report.md` with four supporting figures, and should be folded into this subsection.

E Parameter Stability

A A cointegration test that fails should be diagnosed, not merely reported. This appendix sets out how instability is tested for in this chapter and what constraints govern reading a failure as evidence of regime change rather than as evidence of no long-run relation.

E.1 What the Shin statistic already tells us

A Shin's C statistic is Hansen's L_c restricted to intercept-only parameter variation, with $L_c^{\text{intercept}} = T \cdot C_{\text{Shin}}$. This identity holds to machine precision across the Specification A models.

B The consequence is interpretive and it matters. A Shin rejection is already evidence of parameter instability; it is not evidence of "no cointegration" in some undifferentiated sense. A report that observes a Shin rejection and stops there has not said what it appears to have said. It must either decompose the rejection with the full Hansen L_c, per parameter and jointly, or state plainly that the decomposition was not performed.

C One numerical caveat travels with this identity wherever both statistics appear: L_c and C_{Shin} will not agree numerically, because they use different long-run variance estimators — a score-process Bartlett estimator against the conditional $\omega_{u \cdot v}$ from the cointegrating regression. On this data they differ by roughly an order of magnitude. The discrepancy is expected and is not a sign that either is wrong.

E.2 Bootstrapped nulls, and why the direction of the null matters

A Test statistics introduced into this chapter take their inference from a bootstrapped null distribution rather than from hardcoded critical values, unless a properly indexed published table has been digitized in full.

B The construction of that null is where the single most dangerous error in this protocol lives. Tests whose null is cointegration with stable parameters — Hansen L_c, MeanF, SupF, and the CUSUM family — require resampling stationary residuals under a moving-block scheme. Tests whose null is no cointegration — Gregory-Hansen — require cumulating resampled differences into a random walk. Transposing the two data-generating processes silently inverts every p-value while producing output that looks entirely plausible. Block length follows the fitted long-run-variance bandwidth; i.i.d. resampling is not admissible, since it destroys exactly the serial dependence the block scheme exists to preserve.

E.3 Which specification the stability tests run on

A Endogenously selected step dummies are excluded from the stability tests. The CUSUM family and Gregory-Hansen run on the core specification — intercept plus stochastic regressors — because a dummy selected from the same data whose stability is being tested contaminates the test. Hansen L_c is reported both on the full fitted design and restricted to core parameters, and any table reporting it states which.

B The distinction between level shifts and vector shifts governs what may be claimed. Step dummies absorb level shifts only. They cannot absorb a shift in the cointegrating vector itself. A specification that answers detected breaks with step dummies has therefore not addressed slope instability, and is not described as break-robust anywhere in this chapter on that basis.

E.4 When a failure may be read as regime change

A Reading a cointegration failure as evidence of structural regime-switching is permitted only when four conditions hold jointly.

B Admissibility must not be the explanation: the integration-order gate passes for every regressor as actually constructed, lag structure included. Instability must be affirmatively demonstrated with bootstrapped p-values, not inferred from the failure itself. Break dates must be corroborated across the Hansen SupF argmax, the CUSUM supF argmax, Bai-Perron, and Gregory-Hansen — where these converge the reading is strengthened, and where they diverge the divergence is reported as divergence rather than resolved to whichever date suits the argument. And the constructive question must be answered: does a long-run relation appear once the cointegrating vector is allowed to break, by regime-split re-estimation or by Gregory-Hansen? If it does not, the honest verdict is that no stable long-run relation has been identified, and the text says so.

C Describing a cointegration failure as evidence of regime change on the strength of demonstrated instability alone, without the other three conditions, is not available.

E.5 Small samples

A Where a sub-sample falls to 30 observations or fewer, bootstrap p-values are reported alongside asymptotics. Below roughly 25 observations the estimate is flagged as underpowered and carries an explicit reliability caveat rather than a bare verdict. Step dummies that are constant within a sub-sample are dropped, and where this happens the text notes that the sub-sample specification therefore differs from the full-sample one.

B [**PLACEHOLDER**] Stability results are not yet reported here. Required: Hansen MATHPLACEHOLDER0XYZ (full design and core-restricted), MeanF, SupF, CUSUM, and Gregory-Hansen for each preferred specification, each with bootstrapped MATHPLACEHOLDER1XYZ-values under the correctly oriented null; the break-date corroboration table across the four detection methods; and the consolidated verdict summary. Source material is in the archived `reports/US_Section31_Stability_Audit/99_archive/`.

C

[**PLACEHOLDER**] One known defect to resolve before those results are reported: the residual ADF implementation used in the Chilean structural-break chain is not lag-augmented, and an unaugmented Dickey-Fuller regression on serially correlated residuals under-rejects severely. On the Specification A models the difference is decisive — 2 of 18 rejections at the 5% level with zero lagged differences against 16 of 18 under an information-criterion-selected lag order. Any prior reading of cointegration failure as regime-switching that rests on the unaugmented statistic must be revisited before it is repeated here.

References